POSH PARKING QUOTIENTS ARE PERMUTATION MODULES EXACTLY FOR UNTWISTED GENERATORS

JASON STACK, NATHAN WILLIAMS

Editorial note. Complete original mathematical paper75–2652, all nonadjacent generator/seed/restriction proofs, exact classification tables and embedded bibliography retained unchanged. Authors retain responsibility for mathematics; no proof repair, additional assumptions, independent author review, group computation or certification.

Editorial note. Original acknowledgement/AI disclosure2075–2079 retained exactly, including handwritten Mathematica/GAP3 hsop code from2024 and various GPT models for exploration/computation/writing. No selected-core discovery attribution or AI-free status is inferred, and no author-process audit is performed.

Editorial note. Only principal posh-pair classification and permutation-quotient characterization count. Special groups, coprime/Coxeter consequences and construction/table certificates are complete support/context, not separate counts.

Allowed standard prerequisites (editorial).

- Complex reflection groups, homogeneous parameter systems, fake degrees, Galois twists and parking representations. Original irreducibility/rank/faithfulness/common-positive-degree assumptions, exact fake-degree conventions and all original character/polynomiality conditions retained. Original external Orlik-Solomon, Marin-Michel, Ito-Okada and Cherednik inputs retain their precise source statements/citations.. Full author mathematical paper, seed/restriction constructions and nonadjacent classification tables supplied; no independent computation, representation classification or proof reconstruction.

1. Introduction

Let W be a finite complex reflection group acting on a vector space V of rank r (the *reflection representation*), and put $S \coloneqq \mathrm{Sym}(V^*)$ with graded homogeneous components $S = \bigoplus_{p=0}^{\infty} S_p$. The *field of definition* of W is

$$K_W \coloneqq \mathbb{Q}\big(\chi_V(w) : w \in W\big).$$

By a theorem of M. Benard and D. Bessis [6, 8], every irreducible complex W-representation can be realized over K_W; see I. Marin and J. Michel [28, Introduction]. Write $\mathfrak{f}_W$ for the *conductor* of this field, the least positive integer such that $K_W \subseteq \mathbb{Q}(\zeta_{\mathfrak{f}_W})$, where $\zeta_{\mathfrak{f}_W} \coloneqq \exp(2\pi i/\mathfrak{f}_W)$. (For irreducible well-generated W, we have $\mathfrak{f}_W$ divides the *Coxeter number*[1] h.) For $\gcd(k, \mathfrak{f}_W) = 1$, let σ_k be the automorphism $\zeta_{\mathfrak{f}_W} \mapsto \zeta_{\mathfrak{f}_W}^k$. The *Galois twist* $(V^*)^{\sigma_k}$ is obtained by applying σ_k entrywise to a matrix realization of V^* over $\mathbb{Q}(\zeta_{\mathfrak{f}_W})$ (note, however, that σ_k can become trivial when restricted to K_W—for example, for real groups, $V \simeq V^*$ so that complex conjugation is a trivial Galois twist).

1.1. Posh hsops and pairs.

Following D. Armstrong, V. Reiner, and B. Rhoades [2], and Rhoades [37], we consider the following parameter spaces.

Definition 1.1. For a positive integer p, a W-stable *homogeneous system of parameters (hsop)* of degree p is an r-dimensional W-stable subspace $\Theta \subseteq S_p$ whose common zero is the origin.[2] It *carries* U if $\Theta \cong U$ as an ungraded W-module. If the action of W on Θ is faithful, we call Θ *posh*. We call a pair (p, U) *posh* if a posh hsop of degree p carrying U exists.

Date: arxiv:2610.07280v1; CC BY 4.0.

[1] For an irreducible well-generated group, the Coxeter number is $h = (N + N^*)/r$, where N is the number of reflections and N^* is the number of reflecting hyperplanes. Using the same formula for h when W is not well-generated, $\mathfrak{f}_W \mid h$ exactly when $W = G(2s, 2, r)$ with $s > 1$ odd and $r \geq 2$.

[2] Equivalently, $S/(\Theta)$ is finite-dimensional.

Example 1.2. For $m > 1$, let the cyclic group $W = C_m$ act on $V = \mathbb{C}$ by multiplication by ζ_m, so that it acts on $x \in V^*$ by ζ_m^{-1}. For every $p > 0$, $\Theta = \mathbb{C}x^p$ is a W-stable hsop. It is posh exactly when $\gcd(p, m) = 1$, so that $(p, (V^*)^{\sigma_p})$ is posh.

In fact, posh hsops always carry some Galois twist of the reflection representation.

Lemma 1.3. *Let $W \subseteq \mathrm{GL}(V)$ be a nontrivial irreducible complex reflection group of rank r, and let $\Theta \subseteq S_p$ be a W-stable hsop carrying U. Then*

$$\Theta \text{ is posh} \iff U \cong (V^*)^\sigma \text{ for some } \sigma.$$

We caution that non-isomorphic twists can form posh pairs with the same degree.

Example 1.4. The Coxeter group H_3 is defined over $K_{H_3} = \mathbb{Q}(\sqrt{5}) \subseteq \mathbb{Q}(\zeta_5)$, and $\mathfrak{f}_{H_3} = 5$. As a real group, complex conjugation (σ_4) acts trivially to preserve the reflection representation, whereas the nontrivial Galois twist $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ is realized by both σ_2 and σ_3. Both $(5, V^*)$ and $(5, (V^*)^\sigma)$ are posh. (For more information, see [4] and Section 4.4.)

Several families of posh pairs were previously known:

- For S_n, M. Haiman proved that (p, V^*) is posh exactly when $\gcd(p, n) = 1$, by composing equivariant polynomial maps [22]. We review his construction in Section 2.6.
- For Weyl groups and $\gcd(p, h) = 1$, A. Oblomkov and Z. Yun give a geometric realization of the posh pairs (p, V^*) using homogeneous affine Springer fibers [31].
- The rational Cherednik approach builds on I. Gordon's work on diagonal-coinvariant quotients [19] and on Y. Berest–P. Etingof–V. Ginzburg's finite-dimensional polynomial representations and character formulas [7]. For irreducible well-generated complex reflection groups and $\gcd(p, h) = 1$, I. Gordon and S. Griffeth use *rational Cherednik algebras* to prove that $(p, (V^*)^{\sigma_p})$ is posh [20, Theorem 2.11 and Section 2.18]. We review their construction in Section 2.6; for P. Etingof's reducedness argument in the real case, see [2, Section 12].

Our first result classifies all posh pairs. In light of the constructions above, we first give the easily-stated (and expected) specialization of our classification to well-generated complex reflection groups.

Proposition 1.5. *Let W be a well-generated finite irreducible complex reflection group, let $p > 0$ be coprime to its Coxeter number h, and let U be a W-module. Then (p, U) is posh iff $U \simeq (V^*)^{\sigma_p}$.*

Note that Proposition 1.5 does not apply to the two non-isomorphic posh hsops of degree 5 for H_3 from Example 1.4, since 5 is not coprime to the Coxeter number $h = 10$. We now state a complete classification of the degrees p and representations U for which there exists a posh hsop of degree p carrying U (necessarily a Galois twist $(V^*)^\sigma$ by Lemma 1.3). In contrast to Proposition 1.5, the particular twist σ is not necessarily specified by the degree p.

Theorem 1.6. *Let W be a nontrivial irreducible finite complex reflection group of rank r, fix a positive integer p, and fix U a faithful r-dimensional representation of W. Write $a_i := e_i(U)$ and $b_i := e_i(U^*)$ for the fake-degree exponents, and let $d_1, \dots, d_r$ be the invariant degrees.*

The following are equivalent:

(1) *The pair (p, U) is posh.*

(2) *The class function*

$$w \longmapsto \frac{\det(1 - q^p w \mid U)}{\det(1 - qw \mid V^*)}$$

is the character of a finite-dimensional graded W-module.

(3) *Every reflection of W acts on U as a reflection, and both*

$$\mathrm{Cat}^+_{p,U}(q) := \prod_{i=1}^{r} \tfrac{1-q^{p+b_i}}{1-q^{d_i}} \qquad \text{and} \qquad \mathrm{Cat}^-_{p,U}(q) := \prod_{i=1}^{r} \tfrac{q^{a_i}-q^p}{1-q^{d_i}}$$

are polynomials.

(4) *The pair (p, U) appears in Table 3 (infinite families) or Table 5 (exceptional types).*

The Koszul character formula of Proposition 2.3 proves (1) ⇒ (2); see D. Bessis and V. Reiner [10, Proposition 4.2 and its proof]. For (2) ⇒ (3), polynomiality of the determinant quotient at each reflection forces it to act on U as a reflection. A theorem of Marin–Michel [28, Theorem 1.2], recalled in Theorem 2.5, classifies faithful irreducible reflection representations as Galois twists. Its extension to reducible modules in Proposition 2.6 then identifies U as a Galois twist of V^*, as shown in Proposition 4.1. The exterior-invariant theorem of P. Orlik and L. Solomon [33, Theorem 3.1], recalled in Theorem 2.4, identifies the

trivial and determinant multiplicities with $\mathrm{Cat}^+_{p,U}$ and $\mathrm{Cat}^-_{p,U}$ in Corollary 3.7. For an actual hsop, these are the Hilbert series of A_Θ^W and $\mathrm{Hom}_W(\det\Theta, A_\Theta)$ in Theorem 3.4. We then identify all pairs satisfying (3) and prove that each is posh by constructing an hsop.

For the infinite families, coordinate powers supply both the hsops and the fake-degree calculation used to prove necessity in Propositions 4.2 and 4.3; Haiman's construction in Example 2.11 handles type A. For the exceptional groups, a finite cyclotomic test lists the candidates in Proposition 4.4. Gordon–Griffeth's construction covers the well-generated coprime cases in Theorem 2.8. The remaining well-generated cases reduce by composition to eight seed hsops for G_4, G_6, H_3, G_{27}; we prove this coverage in Proposition 4.6 and verify the seed maps in Lemma 4.5. Scalar extensions and restriction from E_8 to G_{31} complete the non-well-generated cases in Lemmas 4.8 and 4.9.

Two operations recur in our constructions:

- composition of posh hsops multiplies their common degrees and composes their Galois twists in Lemma 2.9; and
- pairing an hsop carrying U with basic U-valued covariants produces invariant parameters of degrees $p + e_i(U^*)$; for $U = V^*$, these are obtained by contracting the hsop map with the differentials of basic invariants in Proposition 3.2.

1.2. **Posh parking spaces.** We now turn to the combinatorics of the (finite-dimensional) quotient of S by a posh hsop.

Definition 1.7. Given an hsop $\Theta \subseteq S_p$, define

$$A_\Theta := S/(\Theta).$$

Here (Θ) denotes the (homogeneous and W-stable) ideal of S generated by the elements of Θ. When Θ is posh, we call A_Θ the corresponding *posh parking space* with parameter p, or simply a *parking space*.

This quotient has vector-space dimension p^r and has the structure of a graded W-module; see Proposition 2.3. We say that a posh hsop is *untwisted* when it carries V^*; in this case the ungraded character of the parking space is $w \mapsto p^{\dim V^w}$ by Eq. (6). In particular, for Weyl groups at all degrees p, and for well-generated complex reflection groups at the Fuss degrees $p = mh + 1$, the posh hsop is untwisted.

When A_Θ happens to be a permutation module, it is interesting to construct combinatorial models for the corresponding ungraded permutation module (similarly to how the group algebra of W is an ungraded combinatorial model of the coinvariant algebra; see Section 2.1 and Problem 5.1). Such combinatorial p-parking modules have been constructed at many levels of generality:

- For the symmetric group S_n, P. H. Edelman [15] gave combinatorial evidence that noncrossing partitions could be used to construct $(h+1)$-parking spaces. D. Armstrong–N. A. Loehr–G. S. Warrington constructed rational parking functions with character $w \mapsto p^{\dim V^w}$ whenever $\gcd(n,p) = 1$ [1, Theorem 3 and Corollary 4].
- Haiman showed that, after forgetting the grading, p-parking spaces for S_n have the permutation character of the finite torus Q/pQ [22]. For *very good* parameters of a Weyl group, E. Sommers [39, Proposition 3.9] showed that Q/pQ has character $w \mapsto p^{\dim V^w}$.
- For Coxeter groups, D. Armstrong–V. Reiner–B. Rhoades [2] and Rhoades [37] constructed noncrossing (and nonnesting) p-parking spaces for $p = h + 1$ and $p = mh + 1$, respectively.
- T. Douvropoulos and M. Josuat-Vergès completed the character formula for noncrossing $(mh+1)$-parking spaces in types E_7 and E_8, and established its analogue for prime noncrossing $(mh-1)$-parking spaces for irreducible Coxeter groups [14, Theorems 3.10 and 3.12].
- J. Stack [40] generalized these constructions for noncrossing $(mh+1)$-parking spaces to well-generated complex reflection groups.

Using character theory, Y. Ito and S. Okada [24, Theorem 1.4] classified exactly when the untwisted determinant quotient is the character of a graded W-module whose underlying ungraded module is a permutation representation. Our second main result resolves a question of Ito–Okada [24, Problem 7.3(2)], since Theorem 1.6 specializes to an algebraic construction of these graded modules as posh parking spaces.

Theorem 1.8. *Let W be a finite irreducible complex reflection group and Θ a posh hsop. Then A_Θ is a permutation posh parking space iff Θ carries V^*.*

The classification is given in Tables 4 and 6. For non-dihedral W, these are also exactly the degrees in which $w \mapsto p^{\dim V^w}$ is a permutation character. For the non-dihedral members of the infinite family $G(m,d,r)$, an untwisted parking space with parameter p exists iff $p \equiv 1 \pmod m$.

Just as in Proposition 1.5, for well-generated groups and degrees coprime to the Coxeter number, we obtain an easily-stated specialization.

Corollary 1.9. *Let W be an irreducible well-generated complex reflection group with Coxeter element*[3] *c of order h, and let Θ be a posh hsop of degree p with $\gcd(p,h)=1$. Write $e_i := d_i - 1$ for the exponents of W.*

The following are equivalent:

- *Θ carries V^*;*
- *A_Θ is a permutation posh parking space;*
- *c^p is conjugate to c in W; and*
- *$\{pe_i \bmod h\}_{i=1}^r = \{e_i\}_{i=1}^r$ as multisets, where residues are taken in $\{0,\dots,h-1\}$.*

Remark 1.10. P. Galashin–T. Lam–M.-T. Trinh–N. Williams constructed rational noncrossing parking objects for finite Coxeter groups and proved that there are p^r such objects when $\gcd(p,h)=1$ [17, Theorem 1.8]. Realizing these objects as an ungraded parking module also requires a W-action with the desired character. For well-generated groups and p coprime to h, we plan to construct combinatorial models for ungraded permutation posh parking spaces, generalizing our work in [12] where we used Springer's theorem that c^p and c are always conjugate in a Weyl group to construct and explain reflection subword models for rational Catalan objects.

1.3. **Outline.** We give background in Section 2. In Section 3 we establish the generator restriction and multiplicity formulas, and prove the permutation criterion Theorem 1.8 using a specialization of the character formula Eq. (6). The classification proof in Section 4 proceeds through the infinite families, the exceptional polynomiality test, the seed constructions, and the restriction constructions, then assembles these results and deduces the coprime specialization. In Section 5 we ask which nonfaithful parameter modules occur and discuss constructions with generators of different degrees. The full classification tables appear in Appendix A.

2. Background

Recall that $S := \mathbb{C}[V] = \mathrm{Sym}(V^*)$ is the polynomial ring on V. In this section, we use the coinvariant algebra to locate copies of a prescribed W-module in S, and explain how an hsop determines the character of its quotient. We then collect the representation-theoretic results and earlier constructions used to prove Theorem 1.6.

2.1. **Coinvariants and parameter spaces.** All representations are over $\mathbb{C}$. A *complex reflection* is a finite-order linear transformation whose fixed space is a hyperplane. A *complex reflection group* is a finite subgroup of $\mathrm{GL}(V)$ generated by complex reflections. A rank-r group is *well-generated* if it can be generated by r reflections. We let W act on S by $(wf)(x) := f(w^{-1}x)$, and write $[p]_q := 1+q+\cdots+q^{p-1}$.

The invariant ring is a polynomial algebra $S^W = \mathbb{C}[f_1,\dots,f_r]$ on homogeneous *basic invariants*; this is the theorem of G. C. Shephard and J. A. Todd [38] and C. Chevalley [11]. Their degrees $d_i := \deg f_i$ are the *invariant degrees*. The *coinvariant algebra* is

$$C_W := S/(f_1,\dots,f_r),$$

where $(f_1,\dots,f_r)$ is the ideal of S generated by the positive-degree homogeneous W-invariants. Chevalley [11] proves that C_W affords the regular representation of W. Choosing graded W-equivariant lifts of C_W to S and multiplying by invariants gives an isomorphism of graded S^W-modules and W-modules

$$S \cong S^W \otimes C_W, \tag{1}$$

with W acting trivially on the first factor. In particular,

$$\mathrm{Hilb}(S^W;q) = \prod_{i=1}^r \frac{1}{1-q^{d_i}}, \qquad \mathrm{Hilb}(C_W;q) = \prod_{i=1}^r [d_i]_q, \qquad \dim C_W = \prod_{i=1}^r d_i = |W|.$$

For a W-module M, its *fake degrees* $e_1(M),\dots,e_{\dim M}(M)$ record its graded multiplicities in C_W:

$$\sum_{j\ge 0} \dim \mathrm{Hom}_W(M,(C_W)_j) q^j = \sum_{i=1}^{\dim M} q^{e_i(M)}.$$

The regular representation ensures that this list has $\dim M$ entries, counted with multiplicity. In particular, $e_i(V) = d_i - 1$.

[3] By *Coxeter element*, we mean a ζ_h-regular element, where $\zeta_h = \exp(2\pi i/h)$; thus $cv = \zeta_h v$ for some v outside every reflecting hyperplane. See [36, Section 1.1].

For an irreducible module U, the following formula tells us in which degrees to look for hsops carrying U, by determining when U occurs in S_p. Such a copy must still have only the origin as its common zero.

Proposition 2.1. *For a W-module U, put $m_U(p) \coloneqq \dim \mathrm{Hom}_W(U, S_p)$. Then*

$$\sum_{p\geq 0} m_U(p)q^p = \frac{\sum_{i=1}^{\dim U} q^{e_i(U)}}{\prod_{i=1}^{r}(1-q^{d_i})}.$$

Proof. Apply $\mathrm{Hom}_W(U,-)$ to Eq. (1). The fake degrees give the multiplicity series in C_W, and the invariant factor has Hilbert series $\prod_i (1-q^{d_i})^{-1}$. □

Example 2.2. Continue Example 1.4 from the introduction, and let $(V^*)^\sigma$ be the nontrivial Galois twist of V^*. The invariant degrees are $2, 6, 10$, whereas the two fake-degree lists are $1, 5, 9$ and $3, 5, 7$. Thus

$$\sum_{p\geq 0} m_{V^*}(p)q^p = \frac{q+q^5+q^9}{(1-q^2)(1-q^6)(1-q^{10})} = q + q^3 + 2q^5 + 3q^7 + 4q^9 + 6q^{11} + O(q^{13}),$$

$$\sum_{p\geq 0} m_{(V^*)^\sigma}(p)q^p = \frac{q^3+q^5+q^7}{(1-q^2)(1-q^6)(1-q^{10})} = q^3 + 2q^5 + 3q^7 + 4q^9 + 5q^{11} + O(q^{13}).$$

Both modules occur with multiplicity two in degree five; we construct hsops of both types in Lemma 4.5. The common-zero condition still needs checking: the unique copy of V^* in degree three is QV^*, where Q is a quadratic invariant, and it vanishes on the nonzero points of $Q = 0$. On the other hand, the unique twisted copy in degree three is an hsop by Theorem 2.8, since 3 is coprime to 10 and σ_3 restricts to the nontrivial automorphism of K_{H_3}.

2.2. **The quotient character.** The equivariant Koszul calculation gives the graded character of the quotient $A_\Theta = S/(\Theta)$ from Definition 1.7; see Bessis–Reiner [10, Proposition 4.2 and its proof]. We recall the argument for a W-stable hsop Θ. This formula gives (1) ⇒ (2) in Theorem 1.6; we also use it to calculate invariant multiplicities in Theorem 3.4 and the ungraded parking character in Theorem 1.8.

Let $\Theta \subseteq S_p$ be a W-stable hsop. Evaluation defines a homogeneous equivariant map

$$V \longrightarrow \Theta^*, \qquad v \longmapsto \big(\theta \longmapsto \theta(v)\big).$$

Its zero fibre is the common zero set of Θ, which is the origin. Equivalently, A_Θ is finite-dimensional. The evaluation map is finite: homogeneous lifts of a basis of A_Θ generate S over the subalgebra $\mathbb{C}[\Theta]$ by induction on degree. Conversely, a finite evaluation map has a finite zero fibre; homogeneity forces this fibre to be the origin. A basis $\theta_1, \dots, \theta_r$ of Θ is a regular sequence, meaning that θ_i is a nonzero-divisor modulo its predecessors. Indeed, S is Cohen–Macaulay and these r elements form a system of parameters [43, Lemmas 10.104.2 and 10.104.7].

Proposition 2.3. *Let $\Theta \subseteq S_p$ be a W-stable hsop. Then*

$$\mathrm{Hilb}(A_\Theta; q) = [p]_q^r, \qquad \chi_{A_\Theta}(w; q) = \frac{\det(1 - q^p w \mid \Theta)}{\det(1 - qw \mid V^*)}.$$

Proof. The Koszul complex has terms $K_j \coloneqq S \otimes \bigwedge^j \Theta$, with each element of Θ in its original degree p, and differential

$$d_j(f \otimes \theta_1 \wedge \cdots \wedge \theta_j) = \sum_{a=1}^{j} (-1)^{a-1} f\theta_a \otimes \theta_1 \wedge \cdots \widehat{\theta_a} \cdots \wedge \theta_j.$$

The differential preserves total degree. The action on K_j is diagonal, and $w(f\theta_a) = (wf)(w\theta_a)$ shows that d_j commutes with W. Regularity makes this complex exact in positive degrees [43, Lemma 15.31.2], giving the exact sequence of graded W-modules

$$0 \longrightarrow K_r \longrightarrow \cdots \longrightarrow K_1 \longrightarrow S \longrightarrow A_\Theta \longrightarrow 0.$$

Its alternating graded character is

$$\chi_{A_\Theta}(w; q) = \chi_S(w; q) \sum_{j=0}^{r} (-q^p)^j \chi_{\wedge^j \Theta}(w).$$

Diagonalizing w gives $\chi_S(w; q) = \det(1 - qw \mid V^*)^{-1}$ and identifies the sum with $\det(1 - q^p w \mid \Theta)$. At $w = 1$, the quotient is $(1-q^p)^r/(1-q)^r = [p]_q^r$. □

2.3. **Exterior invariants.** The character of A_Θ in Proposition 2.3 is an alternating sum of the characters of $S \otimes \bigwedge^j \Theta$. To compute its determinant multiplicity in Theorem 3.4, we use the exterior-algebra theorem of Orlik–Solomon [33, Theorem 3.1] in the following form (see also [3, Theorem 2.7]).

Recall that a W-module M is *amenable* if $\sum_i e_i(M) = e_1(\det M)$, where $\det M \coloneqq \bigwedge^{\dim M} M$ is its one-dimensional determinant representation.

Theorem 2.4 (Orlik–Solomon [33, Theorem 3.1])**.** *If M is amenable, then $(S \otimes \bigwedge M^*)^W$ is an exterior algebra over S^W on $\dim M$ generators with bidegrees $(e_i(M), 1)$. Consequently,*

$$\operatorname{Hilb}\big((S \otimes \textstyle\bigwedge M^*)^W; q, u\big) = \dfrac{\prod_{i=1}^{\dim M}(1+uq^{e_i(M)})}{\prod_{i=1}^{r}(1-q^{d_i})},$$

where q records polynomial degree and u records exterior degree.

2.4. **Galois twists.** As in the introduction, let $K_W \coloneqq \mathbb{Q}(\chi_V(w) : w \in W)$. Its *conductor* $\mathfrak{f}_W$ is the least positive integer such that $K_W \subseteq \mathbb{Q}(\zeta_{\mathfrak{f}_W})$. Every irreducible W-module can be realized over K_W by Benard–Bessis [6, 8]; see also Marin–Michel [28, Introduction]. For $\gcd(k, \mathfrak{f}_W) = 1$, let σ_k send $\zeta_{\mathfrak{f}_W}$ to $\zeta_{\mathfrak{f}_W}^k$. Applying it to a matrix realization defines the *Galois twist* M^{σ_k}; equivalently,

$$\chi_{M^{\sigma_k}}(w) = \sigma_k(\chi_M(w)).$$

Only the restriction to K_W affects the isomorphism class, so different k can specify the same twist. The notation σ_p requires $\gcd(p, \mathfrak{f}_W) = 1$; otherwise the twist must be specified independently of the degree. The conductor $\mathfrak{f}_W$ measures the cyclotomic field needed to define the twists; the degree period M records when a fixed twist occurs as an hsop. For example, $K_{H_3} = \mathbb{Q}(\sqrt{5})$ has conductor 5, whereas its hsop degrees have period 10 in Table 5: the degree conditions also exclude even integers.

When Θ carries $(V^*)^\sigma$, its evaluation map has target $\Theta^* \cong V^\sigma$. This accounts for the two fake-degree lists in Theorem 1.6. For polynomial maps we extend σ to an automorphism of $\mathbb{C}$ before applying it to coefficients.

Theorem 2.5 (Marin–Michel [28, Theorem 1.2])**.** *Let $W \subseteq \mathrm{GL}(V)$ be an irreducible complex reflection group. If M is a faithful irreducible W-module on which every reflection of W acts as a reflection, then $M \cong V^\sigma$ for some $\sigma \in \mathrm{Gal}(K_W/\mathbb{Q})$.*

The irreducibility hypothesis on the module can be removed by allowing a fixed summand. We will apply the following consequence to hsop generators in Lemma 1.3.

Proposition 2.6. *Let $W \subseteq \mathrm{GL}(V)$ be a nontrivial irreducible complex reflection group. If M is a faithful W-module on which every reflection of W acts as a reflection, then*

$$M \cong M^W \oplus V^\sigma$$

for some $\sigma \in \mathrm{Gal}(K_W/\mathbb{Q})$.

Proof. Write $M = M^W \oplus M_1 \oplus \cdots \oplus M_k$ with the M_i nontrivial and irreducible. Each reflection acts nontrivially on exactly one M_i. Let W_i be generated by the reflections assigned to M_i. Each W_i acts trivially on M^W and on all M_j with $j \neq i$. Thus distinct factors commute on M, and any element of $W_i \cap \prod_{j \neq i} W_j$ acts trivially on every summand of M. Faithfulness implies that the W_i commute in W and that these intersections are trivial. Since the reflections generate W, we have $W = W_1 \times \cdots \times W_k$. Write the irreducible reflection representation as $V = V_1 \boxtimes \cdots \boxtimes V_k$. For a generating reflection $s \in W_i$,

$$1 = \operatorname{rank}(s - 1 \mid V_i) \prod_{j \neq i} \dim V_j.$$

If $k > 1$, these identities force every V_i to be one-dimensional. But then every nonidentity element of W is a reflection, whereas a product of nonidentity elements from two factors acts nontrivially on two summands of M. Therefore $k = 1$, and Theorem 2.5 applies to M_1. □

2.5. **Untwisted parking characters.** Ito–Okada [24, Theorem 1.4] classify the degrees for which the untwisted determinant quotient is a graded character, and relate them to the permutation character $w \mapsto p^{\dim V^w}$. Their theorem, recalled below, supplies the untwisted implication in Theorem 1.8. Our proof of Theorem 1.6 realizes these graded characters as quotients by posh hsops of degree p.

Theorem 2.7 (Ito–Okada [24, Theorem 1.4])**.** *Let W be an irreducible finite complex reflection group and p a positive integer. The class function*

$$w \longmapsto \frac{\det(1 - q^p w \mid V^*)}{\det(1 - qw \mid V^*)}$$

is the character of a finite-dimensional graded W-module if and only if both $\mathrm{Cat}^+_{p,V^*}(q)$ *and* $\mathrm{Cat}^-_{p,V^*}(q)$ *from Theorem 1.6 are polynomials. Whenever these conditions hold,* $w \mapsto p^{\dim V^w}$ *is a permutation character. For non-dihedral* W*, the converse also holds.*

Proof. Ito–Okada [24, Theorem 1.4(A)] use V in their determinant quotient. Replacing w by w^{-1} gives our quotient and dualizes each homogeneous representation. Specializing at $q = 1$ gives $w \mapsto p^{\dim V^w}$. Parts (B) and (C) of their theorem imply that whenever this is a character, it is a permutation character. Part (B) gives the converse outside dihedral type. □

2.6. **Earlier hsop constructions.** We review the Gordon–Griffeth construction supplying the well-generated coprime cases, and the composition that extends exceptional seed maps to further degrees. As an application of this composition, we review Haiman's construction in type A in Example 2.11.

Theorem 2.8 (Gordon–Griffeth [20, Theorem 2.11 and Section 2.18])**.** *Let* W *be irreducible and well-generated, with Coxeter number* $h = d_r$*, and let* $p > 0$ *be relatively prime to* h*. Then* $(p, (V^*)^{\sigma_p})$ *is posh, where* $\sigma_p(\zeta_h) = \zeta_h^p$*.*

Construction. BMR freeness in characteristic zero makes the generic Hecke algebra of W free of rank $|W|$ over its parameter ring, supplying the freeness hypothesis in Gordon–Griffeth's construction. See Etingof [16, Theorem 1.3] for the deduction from results of Marin [26, 27] and Marin–Pfeiffer [29, Theorem 1.2].

In the sign convention of Gordon–Griffeth [20, Sections 2.9–2.11], the polynomial representation of the rational Cherednik algebra at parameter $-1/h$ has the trivial module as a quotient, resolved by its coordinate functions. Their equivalence of categories transports this resolution to parameter $-p/h$. Writing $\Delta(M)$ for the standard module, whose underlying S-module is $S \otimes M$, the last part is

$$\Delta((V^*)^{\sigma_p}) \longrightarrow \Delta(\mathbf{1}) = S \longrightarrow L \longrightarrow 0.$$

The singular generating copy of $(V^*)^{\sigma_p}$ maps into S_p and generates the kernel. Since L is finite-dimensional, its image is an hsop. Thus the construction supplies a parameter space inside S without specifying polynomial coordinates. □

Lemma 2.9. *If* $(a, (V^*)^{\sigma})$ *and* $(b, (V^*)^{\tau})$ *are posh, then so is* $(ab, (V^*)^{\sigma\tau})$*.*

Proof. Choose corresponding hsop maps $F : V \to V^{\sigma}$ and $G : V \to V^{\tau}$. Applying σ to the coefficients and covariance equations of G gives $G^{\sigma} : V^{\sigma} \to V^{\sigma\tau}$. The composite $G^{\sigma} \circ F$ is equivariant and homogeneous of degree ab. Its zero fibre is $F^{-1}((G^{\sigma})^{-1}(0)) = F^{-1}(0) = \{0\}$. Galois actions on the character field are abelian, so its representation type depends only on the two twist classes. □

Definition 2.10. A *seed hsop* is a posh hsop used with Lemma 2.9 to construct posh hsops in further degrees. We will give exceptional seed hsops in Section 4.4 and use them in Section 4.4.1 to obtain the remaining posh pairs for G_4, G_6, H_3, G_{27}.

Example 2.11. Let $W = S_n$ and $V = \{x \in \mathbb{C}^n : \sum_i x_i = 0\}$. For a prime $\ell \nmid n$, Haiman [22, Proposition 2.5.4] uses the projected power map

$$H_\ell(x)_i \coloneqq x_i^\ell - \frac{1}{n}\sum_j x_j^\ell. \tag{2}$$

It is homogeneous, S_n-equivariant, and takes values in V. If $H_\ell(x) = 0$, all x_i^ℓ agree. A nonzero zero can therefore be rescaled so that every x_i is an ℓth root of unity. Writing $x_i = \zeta_\ell^{a_i}$ gives $\sum_i \zeta_\ell^{a_i} = 0$. The polynomial $\sum_i t^{a_i}$ is divisible by $1 + t + \cdots + t^{\ell-1}$; evaluation at $t = 1$ forces $\ell \mid n$, a contradiction. Factoring p into primes and composing these maps proves that (p, V^*) is posh whenever $\gcd(p, n) = 1$.

3. Generator representations and parking characters

For irreducible W, we identify the generator modules of posh hsops as Galois twists of V^* in Lemma 1.3 and compute both Catalan multiplicities in Theorem 3.4. We then determine which parking spaces are permutation modules in Theorem 1.8. These restrictions are used to classify posh pairs in Section 4.

3.1. **Generator restriction and multiplicities.** We first give a local criterion for applying the exterior-invariant theorem of Theorem 2.4. The hsop condition will force this criterion to hold, allowing us to compute both multiplicities and identify the faithful generator modules.

For a reflecting hyperplane H, let $W_H \coloneqq \{w \in W : w|_H = 1\}$ be its cyclic pointwise stabilizer, and put $e_H \coloneqq |W_H|$. Choose its generator s_H so that its eigenvalue on V/H is $\zeta_H \coloneqq \exp(2\pi i/e_H)$. The local exponent formula of E. A. Gutkin and E. M. Opdam [32, Lemma 2.1] and V. Beck's criterion [5, Definition 4(ii) and Remark 1(i)] give the following characterization.

Proposition 3.1 ([32, Lemma 2.1], [5, Remark 1(i)])**.** *For a finite-dimensional W-module M, the following are equivalent:*

(1) *both M and M^* are amenable;*
(2) *every reflection s satisfies* $\operatorname{rank}(1 - s \mid M) \le 1$*;*
(3) *every reflection acts on M as a reflection or the identity.*

Proof. Write

$$m_{H,j} \coloneqq \dim \ker(s_H - \zeta_H^{-j} I \mid M), \qquad 0 \le j < e_H,$$

so $m_{H,j}$ is the multiplicity of the indicated eigenvalue, and put

$$J_H(M) \coloneqq \sum_{j=1}^{e_H - 1} j m_{H,j}, \qquad a_H(M) \coloneqq \sum_{j=1}^{e_H - 1} m_{H,j}.$$

Let $\overline{J}_H(M)$ be the least nonnegative residue of $J_H(M)$ modulo e_H. The sum formula of Gutkin–Opdam [32, Lemma 2.1],

$$\sum_i e_i(M) = \sum_H J_H(M),$$

extends from irreducible modules to arbitrary M by additivity. Applying its one-dimensional case to $\det M$ gives $e_1(\det M) = \sum_H \overline{J}_H(M)$, and therefore

$$\sum_i e_i(M) - e_1(\det M) = \sum_H \big(J_H(M) - \overline{J}_H(M)\big).$$

Every summand on the right is nonnegative, so M is amenable exactly when $J_H(M) < e_H$ for every H. Also

$$J_H(M^*) = e_H a_H(M) - J_H(M).$$

Consequently both modules are amenable exactly when $a_H(M) \le 1$ for every H: if both local sums are less than e_H, their sum $e_H a_H(M)$ is less than $2e_H$; conversely, at most one nontrivial eigenvalue makes both sums less than e_H. This is Beck's local criterion [5, Remark 1(i)], and is equivalent to (2) and (3). □

Multiplication sends basic covariants to generators of $(\Theta) \cap S^W$, explaining the numerator degrees in the invariant Hilbert series. These invariant parameters also provide examples in Section 5.

Proposition 3.2. *Let $\Theta \subseteq S_p$ be a W-stable hsop. The ideal $(\Theta) \cap S^W$ is generated by a regular sequence of degrees $p + e_i(\Theta^*)$, so*

$$\operatorname{Hilb}(A_\Theta^W; q) = \prod_{i=1}^{r} \frac{[p + e_i(\Theta^*)]_q}{[d_i]_q}. \tag{3}$$

Proof. Multiplication gives the W-equivariant presentation

$$S \otimes \Theta \xrightarrow{\mu} S \longrightarrow A_\Theta \longrightarrow 0, \qquad \mu(f \otimes \theta) = f\theta.$$

Taking W-invariants is exact, so

$$(S \otimes \Theta)^W \xrightarrow{\mu} S^W \longrightarrow A_\Theta^W \longrightarrow 0.$$

By Eq. (1), $(S \otimes \Theta)^W$ is a free S^W-module of rank r. Its homogeneous basis $c_1, \dots, c_r$ has total degrees $p + e_i(\Theta^*)$: the polynomial coefficients have degrees $e_i(\Theta^*)$, and Θ has degree p. Such invariant tensors are called *covariants*; equivalently, their polynomial coefficients define equivariant maps $V \to \Theta$. Set

$$g_i \coloneqq \mu(c_i), \qquad \deg g_i = p + e_i(\Theta^*).$$

The image of μ is $(\Theta) \cap S^W$, so

$$A_\Theta^W \cong S^W/(g_1, \dots, g_r).$$

This quotient is finite-dimensional. Thus $g_1, \dots, g_r$ form a system of parameters in the polynomial ring S^W and therefore a regular sequence [43, Lemmas 10.104.2 and 10.104.7]. Their degrees give the stated Hilbert series. □

Example 3.3. For an untwisted hsop, choose a W-isomorphism $V^* \cong \Theta$ taking a coordinate basis $x_1, \dots, x_r$ to $\theta_1, \dots, \theta_r$. The theorem of Solomon [3, Theorem 2.3] on invariant differential forms gives the covariant basis $df_1, \dots, df_r$ for homogeneous basic invariants f_i. The invariant parameters are

$$g_i = \sum_{j=1}^{r} \frac{\partial f_i}{\partial x_j}\theta_j, \qquad \deg g_i = p + d_i - 1.$$

For well-generated W and $p = mh + 1$ with $m \geq 0$ an integer, the Hilbert series becomes the Fuss q-Catalan product $\prod_i [mh + d_i]_q / [d_i]_q$.

If Θ carries $(V^*)^\sigma$, then $e_i(\Theta^*) = e_i(V^\sigma)$. In this case the Hilbert series is the formula of Bessis–Reiner [10, Proposition 4.2].

We now show that the hsop condition forces the local reflection criterion in Proposition 3.1, and compute the determinant multiplicity alongside the invariant Hilbert series.

Theorem 3.4. *Let $W \subseteq \mathrm{GL}(V)$ be a finite complex reflection group of rank r, and let $\Theta \subseteq S_p$, $p > 0$, be a W-stable hsop. Every reflection acts on Θ as a reflection or the identity, and both Θ and Θ^* are amenable. Moreover,*

$$\mathrm{Hilb}(A_\Theta^W; q) = \prod_{i=1}^{r} \frac{1 - q^{p+e_i(\Theta^*)}}{1 - q^{d_i}}, \tag{4}$$

$$\mathrm{Hilb}\big(\mathrm{Hom}_W(\det \Theta, A_\Theta); q\big) = \prod_{i=1}^{r} \frac{q^{e_i(\Theta)} - q^p}{1 - q^{d_i}}. \tag{5}$$

Both products are polynomials with nonnegative integer coefficients. At $q = 1$, the first gives $\dim A_\Theta^W = |W|^{-1} \prod_{i=1}^{r} (p + e_i(\Theta^*))$.

Proof. Let s be a reflection with reflecting hyperplane H, and choose an eigenbasis of Θ for s. If $s\theta = \lambda\theta$ with $\lambda \neq 1$, then for $v \in H$ we have $\lambda\theta(v) = (s\theta)(v) = \theta(s^{-1}v) = \theta(v)$, so $\theta|_H = 0$. Thus the restrictions of the $\dim \Theta^s$ fixed basis elements already have the origin as their common zero, because Θ is an hsop. The radical of the ideal they generate in $\mathbb{C}[H]$ is therefore the maximal ideal of the origin, of height $r - 1$. Krull's height theorem [43, Lemma 10.60.12(1)] gives $r - 1 \leq \dim \Theta^s$. Since $\dim \Theta = r$, every reflection acts on Θ as a reflection or the identity. By Proposition 3.1, both Θ and Θ^* are amenable.

The invariant product of Eq. (4) follows directly from Proposition 3.2. For the determinant multiplicity, we follow Bessis–Reiner [10, Proposition 4.2], using the exterior-algebra theorem of Orlik–Solomon [33, Theorem 3.1], recalled in Theorem 2.4. The natural pairing of complementary exterior powers gives $\bigwedge^j \Theta \otimes (\det \Theta)^* \cong \bigwedge^{r-j} \Theta^*$. Tensoring the Koszul resolution in Proposition 2.3 with the ungraded representation $(\det \Theta)^*$ and taking W-invariants computes $\mathrm{Hom}_W(\det \Theta, A_\Theta)$. Both operations preserve exactness. In the following calculation the exterior powers are regarded as ungraded representations; the factors q^{pj} account for their degrees in the resolution. Applying Theorem 2.4 to Θ gives

$$\begin{aligned} \mathrm{Hilb}\big(\mathrm{Hom}_W(\det \Theta, A_\Theta); q\big) &= \sum_{j=0}^{r} (-q^p)^j \,\mathrm{Hilb}\big((S \otimes \textstyle\bigwedge^{r-j} \Theta^*)^W; q\big) \\ &= \prod_{i=1}^{r} \frac{q^{e_i(\Theta)} - q^p}{1 - q^{d_i}}. \end{aligned}$$

Indeed, the term indexed by j selects $r - j$ exterior generators. Each selected generator contributes $q^{e_i(\Theta)}$, and each of the j omitted generators contributes $-q^p$. Summing over all subsets gives the product. These are Hilbert series of graded multiplicity spaces, so their coefficients are nonnegative integers. Taking $q \to 1$ in Eq. (4) and using $\prod_i d_i = |W|$ proves the dimension formula. □

When Θ carries $U = (V^*)^\sigma$, the two products are $\mathrm{Cat}^+_{p,U}$ and $\mathrm{Cat}^-_{p,U}$ from Theorem 1.6.

Lemma 1.3. *Let $W \subseteq \mathrm{GL}(V)$ be a nontrivial irreducible complex reflection group of rank r, and let $\Theta \subseteq S_p$ be a W-stable hsop carrying U. Then*

$$\Theta \text{ is posh} \iff U \cong (V^*)^\sigma \text{ for some } \sigma.$$

Proof. A Galois twist of V^* is faithful. Conversely, suppose that U is faithful. By Theorem 3.4, every reflection acts on U^* as a reflection. Thus Proposition 2.6 gives $U^* \cong (U^*)^W \oplus V^\sigma$. Since $\dim U = r$, the fixed summand vanishes, and dualizing gives $U \cong (V^*)^\sigma$. □

The same conclusion extends to faithful homogeneous regular sequences.

Corollary 3.5. *Let $W \subseteq \mathrm{GL}(V)$ be a nontrivial irreducible complex reflection group of rank r. If $\Theta \subseteq S_p$, $p > 0$, is a faithful W-stable subspace whose basis is a regular sequence, then $\dim \Theta = r$ and Θ is a posh hsop carrying a Galois twist of V^*.*

Proof. Put $\ell := \dim \Theta$. Regularity gives $\ell \le r$ and $\dim Z = r - \ell$ for $Z := V(\Theta)$. For a reflection s with hyperplane H, choose an s-eigenbasis of Θ. Every nonfixed eigenvector vanishes on H, so $Z \cap H$ is cut out inside H by the restrictions of the fixed eigenvectors. Therefore

$$r - 1 - \dim \Theta^s \le \dim(Z \cap H) \le r - \ell,$$

and $\dim \Theta^s \ge \ell - 1$. Thus every reflection acts on Θ^* as a reflection or the identity, and faithfulness excludes the identity. By Proposition 2.6, Θ^* contains a summand isomorphic to V^σ, so $\ell \ge r$. Therefore $\ell = r$ and Z has dimension zero. Since Z is a cone containing the origin, $Z = \{0\}$ and Θ is an hsop. The fixed summand vanishes, and dualizing proves the claim. □

Remark 3.6. Thus Theorem 1.6 classifies the degrees and generator modules of all faithful W-stable regular sequences of common positive degree: each has length r and carries a Galois twist. The nonfaithful parameter problem is discussed in Section 5.

Corollary 3.7. *Let $U = (V^*)^\sigma$. If the determinant quotient in Theorem 1.6(2) is the character of a finite-dimensional graded W-module, then both $\mathrm{Cat}^+_{p,U}$ and $\mathrm{Cat}^-_{p,U}$ belong to $\mathbb{Z}_{\ge 0}[q]$.*

Proof. A Galois twist of a reflection is again a reflection, so Proposition 3.1 proves amenability of U and U^* independently of hsop existence. Expanding the numerator as in Proposition 2.3 gives the same alternating exterior-power series used in the proof of Theorem 3.4. This calculation requires only the determinant quotient, so it applies without an hsop. Taking its trivial and $\det U$ multiplicities and applying Theorem 2.4 gives $\mathrm{Cat}^+_{p,U}$ and $\mathrm{Cat}^-_{p,U}$, respectively. If the quotient is the character of a finite-dimensional graded module, these multiplicities are polynomials with nonnegative integer coefficients. This gives the polynomiality part of (2) ⇒ (3) in Theorem 1.6 once U is known to be a Galois twist. □

3.2. **Untwisted parameters and permutation modules.** We now prove that a parking space is a permutation module exactly when its parameters carry V^*.

The *reduced determinant* $\mathrm{rdet}(1 - w)$ of a finite-order operator w is the product of the factors $1 - \lambda$ with $\lambda \ne 1$. Let Θ be a posh hsop of degree p carrying $U = (V^*)^\sigma$, as in Lemma 1.3. In the character formula of Proposition 2.3, the numerator and denominator have the same number $\dim V^w$ of vanishing factors as $q \to 1$. Each pair contributes p, while Galois conjugation sends the remaining eigenvalues on V^* to those on U. Therefore

$$\chi_{A_\Theta}(w) = p^{\dim V^w} \frac{\sigma\big(\mathrm{rdet}(1 - w \mid V^*)\big)}{\mathrm{rdet}(1 - w \mid V^*)}. \tag{6}$$

This is the hsop character formula in Bessis–Reiner [10, Proposition 4.2, equation (4.2)]. The untwisted specialization $w \mapsto p^{\dim V^w}$ also appears in the Orlik–Solomon expansion into parabolic permutation characters [34, 35]; see Ito–Okada [24, Proposition 2.3]. The exterior-invariant identity of Orlik–Solomon [33, Theorem 3.1] underlies the multiplicity products in Theorem 3.4; here we obtain the character value directly by specializing Proposition 2.3.

Theorem 1.8. *Let W be a finite irreducible complex reflection group and Θ a posh hsop. Then A_Θ is a permutation posh parking space iff Θ carries V^*.*

Proof. Suppose that A_Θ is a permutation module, and write $U = (V^*)^\sigma$. If w is *elliptic*, meaning $V^w = 0$, then Eq. (6) gives

$$\chi_{A_\Theta}(w) = \frac{\sigma\big(\det(1 - w \mid V^*)\big)}{\det(1 - w \mid V^*)}.$$

This is a nonzero permutation-character value, therefore a positive integer. Its norm from K_W to $\mathbb{Q}$ is one, so it equals one. On nonelliptic elements $\det(1 - w \mid V^*) = 0$. Consequently

$$\sigma\big(\det(1 - w \mid V^*)\big) = \det(1 - w \mid V^*) \qquad (w \in W).$$

If $r = 1$, then $\det(1 - w \mid V^*) = 1 - \chi_{V^*}(w)$, so $(V^*)^\sigma \cong V^*$. Suppose $r > 1$. The exterior powers $\bigwedge^j V^*$, $0 \le j \le r$, are irreducible and pairwise nonisomorphic by H. Hu [23, Theorem 1.2]. Therefore

$$\det(1 - w \mid V^*) = \sum_{j=0}^{r} (-1)^j \chi_{\bigwedge^j V^*}(w)$$

is an expansion into distinct irreducible characters. Galois conjugation preserves irreducibility and distinctness. Since this virtual character is fixed by σ, the character of $(V^*)^\sigma$ occurs in the expansion with coefficient -1. Thus $(V^*)^\sigma \cong \bigwedge^j V^*$ for an odd j. Dimensions give $\binom{r}{j} = r$, so $j = 1$ or $j = r - 1$. For $r > 2$, a reflection with nontrivial eigenvalue λ on V^* has eigenvalue λ with multiplicity $r - 1$ on $\bigwedge^{r-1} V^*$. Its fixed space there has codimension $r - 1$, whereas its fixed space on $(V^*)^\sigma$ has codimension one. This excludes $j = r - 1$. For $r = 2$, the two possibilities coincide. Therefore $U \cong V^*$.

Conversely, if $U \cong V^*$, then Eq. (6) gives $\chi_{A_\Theta}(w) = p^{\dim V^w}$. Ito–Okada [24, Theorem 1.4] show that this is a permutation character whenever the untwisted graded character exists, including the dihedral case; see Theorem 2.7. □

Example 3.8. Continue Example 2.2, with $\sigma(\sqrt{5}) = -\sqrt{5}$. We will construct posh hsops of degree 5 in Lemma 4.5 that give quotients of vector-space dimension 125 by Proposition 2.3. By Theorem 3.4, their invariant Hilbert series are

$$\mathrm{Cat}^+_{5,V^*}(q) = \frac{1 - q^{14}}{1 - q^2} = 1 + q^2 + q^4 + q^6 + q^8 + q^{10} + q^{12},$$
$$\mathrm{Cat}^+_{5,(V^*)^\sigma}(q) = \frac{(1 - q^8)(1 - q^{12})}{(1 - q^2)(1 - q^6)} = 1 + q^2 + q^4 + 2q^6 + q^8 + q^{10} + q^{12}.$$

Their invariant dimensions are 7 and 8, respectively. Both determinant multiplicities vanish, since both fake-degree lists contain 5.

View H_3 as the full symmetry group of the icosahedron. Its longest element is the central inversion $w_0 = -I$. Let R be a rotation through $2\pi/5$ about an axis through opposite vertices, and put $w := w_0 R = -R$. Thus w has order ten and $w^5 = w_0$; in particular, w is not the longest element. Its eigenvalues on V^* are $-1, -\zeta_5, -\zeta_5^{-1}$, so w is elliptic and

$$\mathrm{rdet}(1 - w \mid V^*) = 2(1 + \zeta_5)(1 + \zeta_5^{-1}) = 3 + \sqrt{5}.$$

For the twisted quotient, Eq. (6) gives

$$\chi_{A_\Theta}(w) = \frac{3 - \sqrt{5}}{3 + \sqrt{5}} = \frac{7 - 3\sqrt{5}}{2}.$$

This irrational value excludes a permutation module. The untwisted quotient is a permutation module by Theorem 1.8.

There is also a converse for non-dihedral groups: Ito–Okada [24, Theorem 1.4] show that $w \mapsto p^{\dim V^w}$ is a character if and only if it is a permutation character, and these conditions give exactly the untwisted degrees in Tables 4 and 6. For dihedral groups the character and permutation-character conditions are still equivalent, but their range can be larger. For $I_2(m)$, Ito–Okada's permutation range is $p = 1$, or $p \ge m - 1$ with

$$p^2 \equiv 1 \pmod{m} \quad (m \text{ odd}), \qquad p^2 \equiv 1 \pmod{2m} \quad (m \text{ even}).$$

This can exceed the range $p \equiv \pm 1 \pmod{m}$ for untwisted posh hsops.

Example 3.9. For $I_2(15)$ and $p = 19$, the class function $w \mapsto 19^{\dim V^w}$ is a permutation character, but there is no untwisted posh hsop of degree 19. The twisted posh hsop of degree 19 exists, and its quotient has a different, non-permutation character.

The package [41] also records the permutation obstruction for exceptional conjugacy classes, independently of the proof of Theorem 1.8.

4. Classification of posh pairs

4.1. **Reduction and proof strategy.** We reduce to Galois twists, determine the pairs satisfying the polynomiality conditions, and construct hsops to prove they are posh.

Theorem 1.6. *Let W be a nontrivial irreducible finite complex reflection group of rank r, fix a positive integer p, and fix U a faithful r-dimensional representation of W. Write $a_i := e_i(U)$ and $b_i := e_i(U^*)$ for the fake-degree exponents, and let $d_1, \dots, d_r$ be the invariant degrees.*

The following are equivalent:

(1) *The pair (p, U) is posh.*
(2) *The class function*
$$w \longmapsto \frac{\det(1 - q^p w \mid U)}{\det(1 - qw \mid V^*)}$$
is the character of a finite-dimensional graded W-module.
(3) *Every reflection of W acts on U as a reflection, and both*
$$\mathrm{Cat}^+_{p,U}(q) := \prod_{i=1}^{r} \tfrac{1-q^{p+b_i}}{1-q^{d_i}} \qquad \text{and} \qquad \mathrm{Cat}^-_{p,U}(q) := \prod_{i=1}^{r} \tfrac{q^{a_i}-q^p}{1-q^{d_i}}$$
are polynomials.
(4) *The pair (p, U) appears in Table 3 (infinite families) or Table 5 (exceptional types).*

Proposition 4.1. *Under the hypotheses of Theorem 1.6, the implications (1) ⇒ (2) ⇒ (3) hold. Moreover, condition (3) forces $U \cong (V^*)^\sigma$ for some Galois automorphism σ.*

Proof. Proposition 2.3 gives (1) ⇒ (2). Suppose that (2) holds. For a reflection s, the denominator of the determinant quotient vanishes to order $r - 1$ at $q = 1$, while the numerator vanishes to order $\dim U^s$. Polynomiality therefore gives $\dim U^s \geq r - 1$. Since U is faithful and has dimension r, every reflection acts on U as a reflection. By Proposition 2.6, applied to U^*, we obtain $U \cong (V^*)^\sigma$ for some σ. Now Corollary 3.7 gives the two polynomiality conditions, proving (2) ⇒ (3). Under (3), the reflection condition, faithfulness, and $\dim U = r$ likewise force U to be a Galois twist of V^*. □

It remains to prove (3) ⇒ (4) ⇒ (1). Section 4.2 proves both implications for the infinite families. For the exceptional groups, Section 4.3 determines all possible pairs, and Table 1 organizes their constructions. We assemble these results in Section 4.6.

Groups or parameters	Construction
Well-generated groups, $\gcd(p, h) = 1$	Theorem 2.8
G_4, G_6, H_3, G_{27}, remaining degrees	Section 4.4.1
$G_7, G_{11}, G_{12}, G_{13}$,	Lemma 4.8
G_{15}, G_{19}, G_{22}	
G_{31}	Lemma 4.9

TABLE 1. Constructions for the exceptional groups in Theorems 1.6 and 1.8, with $h = d_r$ for a well-generated group. The degree and generator types are listed in Tables 5 and 6.

4.2. **Infinite families.** For $d \mid m$, let $G(m, d, r)$ be the group of $r \times r$ monomial matrices whose nonzero entries are mth roots of unity and whose product is an (m/d)th root of unity. In rank one we take $d = 1$, so $G(m, 1, 1) = C_m$. The symmetric group S_n acts on its rank-$(n-1)$ reflection representation.

Proposition 4.2. *Under the hypotheses of Theorem 1.6, condition (3) holds for the infinite families exactly for the following pairs:*

(1) *$W = S_n$: $\gcd(p, n) = 1$ and $U \cong V^*$;*
(2) *$W = G(m, d, r)$ irreducible, with $m \geq 2$: $\gcd(p, m) = 1$ and $U \cong (V^*)^{\sigma_p}$.*

Every listed pair is posh; the list agrees with Table 3.

We first construct the coordinate-power hsops and use them to compute the fake degrees needed for necessity.

Proposition 4.3. *Let $W = G(m, d, r)$, where $m \geq 2$, $d \mid m$, and $r \geq 2$. For $1 \leq k \leq m$, let M_k be the monomial representation obtained by replacing each nonzero matrix entry λ by λ^k, and put $U_k := M_k^*$. For every $p = mj + k$, $j \geq 0$, the subspace*
$$\Theta_p := \mathrm{span}\{x_1^p, \dots, x_r^p\}$$
is an hsop carrying U_k. Put $a := m/d$ and $c_k := \lceil k/a \rceil$. Then
$$\{e_i(U_k^*)\} = \{m - k, 2m - k, \dots, (r-1)m - k,\ rac_k - k\}. \tag{7}$$
In particular, no coprimality assumption on k is needed.

Proof. Entrywise powers preserve products of monomial matrices, so M_k is a representation. The coordinate-power map has the stated covariance and only the origin as its zero. A monomial is fixed by the diagonal subgroup exactly when all its exponents are congruent to ba modulo m for some $0 \le b < d$. In the range $0, \dots, mj + k - 1$, exactly c_k of these residue classes have $j + 1$ representatives and the others have j. Coordinate permutations therefore give

$$\dim A_{\Theta_p}^W = c_k \binom{j+r}{r} + (d - c_k)\binom{j+r-1}{r} = \frac{(dj + rc_k)(j+1)\cdots(j+r-1)}{r!}.$$

Substituting $p = mj+k$ in the dimension formula of Theorem 3.4 and using $|W| = m^r r!/d$ proves Eq. (7) by equality of monic polynomials for infinitely many j. □

Proof of Proposition 4.2. By Proposition 4.1, it suffices to consider Galois twists of V^*.

Coordinate-power constructions. For C_m and $G(m, d, r)$, write $U = (V^*)^{\sigma_k}$, where $1 \le k < m$, $\gcd(k, m) = 1$, and $\sigma_k(\zeta_m) = \zeta_m^k$. Coordinate powers

$$F_p(x_1, \dots, x_r) \coloneqq (x_1^p, \dots, x_r^p)$$

give the required hsop when $p \equiv k \pmod m$, by Proposition 4.3 and its immediate rank-one analogue. In the dihedral case, swapped coordinate powers give the residue $-k$ as well.

Fake degrees. The exponents $b_i = e_i(U^*)$ are given by Eq. (7), with $a = m/d$ and $c_k = \lceil k/a \rceil$ as in Proposition 4.3. The $a_i = e_i(U)$ are obtained by replacing k with $m - k$ in that formula.

Necessary congruences. If $r \ge 3$, the denominator of $\mathrm{Cat}^+_{p,U}$ has a zero of order at least $r - 1$ at a primitive mth root of unity. If $p \not\equiv k \pmod m$, at most its last numerator factor vanishes, which is impossible. If $r = 2$ and $d < m$, assume again $p \not\equiv k \pmod m$ and set $c' \coloneqq c_{m-k}$. Polynomiality of Cat^+ forces $p - k \equiv -2ac_k \pmod m$, whereas polynomiality of Cat^- forces $p - k \equiv 2ac' \pmod m$. The latter congruence also holds if $\mathrm{Cat}^- = 0$, since its first exponent is k. As $c_k + c' = d + 1$, these congruences force $d \mid 2$. But then both invariant degrees $m, 2m/d$ are divisible by m, and the numerator of Cat^+ has only one vanishing factor, again impossible. For $I_2(m)$, the exponents are $k, m-k$, so polynomiality at a primitive mth root forces $p \equiv \pm k$. The twists indexed by k and $-k$ are isomorphic, so either congruence gives $U \cong (V^*)^{\sigma_p}$. The rank-one assertion is immediate.

Type A. For S_n there is only the trivial twist. The classification recalled in Theorem 2.7 makes (3) equivalent to $\gcd(p, n) = 1$. Haiman's construction in Example 2.11, or Theorem 2.8, supplies the hsop. □

In particular, for the Weyl groups of types B_r and C_r, both realized as $G(2, 1, r)$, and of type $D_r = G(2, 2, r)$, a posh hsop of degree p exists exactly when p is odd, and it is untwisted. For every positive odd p, the coordinate powers $\Theta_p = \mathrm{span}\{x_1^p, \dots, x_r^p\}$ give such an hsop. Thus no coprimality condition with h is required in these types: for example, $p = 3$ is allowed in both B_3 and D_4, whose Coxeter number is $h = 6$.

4.3. **Exceptional polynomiality conditions.**

Proposition 4.4. *Let W be an exceptional irreducible complex reflection group and U a Galois twist of V^*. Both products in Theorem 1.6(3) are polynomials exactly for the pairs (p, U) in Table 5.*

Proof. Finite reduction. For each group and twist put $L \coloneqq \mathrm{lcm}(d_1, \dots, d_r)$ and $A \coloneqq \max_i a_i$. Above A, polynomiality of $\mathrm{Cat}^+_{p,U}(q)$ and $\mathrm{Cat}^-_{p,U}(q)$ depends only on $p \pmod L$: each factor $1 - q^n$ contains the cyclotomic polynomial $\Phi_\delta(q)$ exactly once when $\delta \mid n$. We test one representative greater than A in every residue class, using the inequalities

$$\#\{i : \delta \mid p + b_i\} \ge \#\{i : \delta \mid d_i\}, \qquad \#\{i : \delta \mid p - a_i\} \ge \#\{i : \delta \mid d_i\}$$

for every δ dividing an invariant degree. We also test every $1 \le p \le A$. If $p = a_i$ for some i, the second product is zero; otherwise its cyclotomic multiplicities are obtained from $|p - a_i|$. Thus finitely many cyclotomic divisibility tests determine all positive degrees.

Evaluation. The finite calculation is implemented in the accompanying computation package [41], using the CHEVIE package [18, 30] for reflection-group data. It checks the 128 distinct Galois twists of the 34 exceptional groups and gives precisely the rows of Table 5, with no additional small-degree exceptions. Testing the divisors of L gives the least period of each row. □

4.4. **The four remaining well-generated groups.** Among well-generated exceptional groups, the coprime construction in Theorem 2.8 leaves precisely the degrees $p \equiv 3 \pmod 6$ for G_4, $p \equiv 3, 9 \pmod{12}$ for G_6, $p \equiv 5 \pmod{10}$ for H_3, and $p \equiv 5, 25 \pmod{30}$ for G_{27}, with the generator twists listed in Table 5. We state eight seed hsops, show how they cover every remaining pair by composition, and then verify the seeds.

The need for constructions beyond the coprime Cherednik theorem already appears in H_3 at $p = 5$. M. Balagović and A. Puranik [4, Theorem 10.1] compute that the spherical simple Cherednik module $L_{1/2}(\mathbf{1})$ has dimension 115; a degree-five hsop quotient has dimension 125. Thus the quintic parking modules cannot be identified with that simple module.

4.4.1. *Seeds and coverage.* We use seed hsops in the sense of Definition 2.10.

Lemma 4.5. *The eight pairs $(p, (V^*)^{\sigma_k})$ in Table 2 are posh.*

W	conductor $\mathfrak{f}_W$	degree p	twist representatives k
G_4	3	3	1, 2
G_6	12	3	7, 11
$H_3 = G_{23}$	5	5	1, 2
G_{27}	15	5	2, 11

TABLE 2. The eight seed hsops, carrying $(V^*)^{\sigma_k}$. Each row contains one map for each listed twist.

Proposition 4.6. *For $W = G_4, G_6, H_3, G_{27}$, every pair in Table 5 is realized by composing an hsop from Theorem 2.8 with zero or more seeds from Lemma 4.5.*

Proof assuming Lemma 4.5. The simultaneous polynomiality calculation in Proposition 4.4 gives the following residue lists. Here k specifies the generator twist with conductor $\mathfrak{f}_W$ as in Table 2, and the degree period is the Coxeter number $h = 6, 12, 10, 30$ for G_4, G_6, H_3, G_{27}, respectively. Each entry lists all allowed residues of the positive degree p:

(8)

W	k	p modulo h
G_4	1	1, 3
	2	3, 5
G_6	1	1, 9
	5	5, 9
	7	3, 7
	11	3, 11
H_3	1	1, 5, 9
	2	3, 5, 7
G_{27}	1	1, 19, 25
	2	5, 17, 23
	7	7, 13, 25
	11	5, 11, 29.

The coprime residues are covered by Theorem 2.8. For every other listed pair (p, U) write $p = \ell^a b$, where $\ell = 3$ for G_4, G_6, $\ell = 5$ for H_3, G_{27}, and $\ell \nmid b$. Then $\gcd(b, h) = 1$. By Lemma 2.9, we may compose the degree-b hsop with a suitably twisted seeds from Lemma 4.5.

For completeness, the twist arithmetic is small. For G_4 and H_3 the seed set is the entire two-element twist group. For G_6 the seed set is $\{7, 11\}$ modulo 12; its even positive powers give $\{1, 5\}$ and its odd powers give $\{7, 11\}$. For G_{27} the character is fixed by $\{1, 4\} \subset (\mathbb{Z}/15\mathbb{Z})^\times$. In the quotient twist group, the seed set $\{2, 11\}$ has even positive powers $\{1, 7\}$ and odd powers $\{2, 11\}$. Multiplication by the twist of the degree-b hsop gives exactly Eq. (8). Thus composition treats all degrees, including arbitrarily large powers of 3 or 5. □

Example 4.7. Continue Example 3.8, with $\sigma(\sqrt{5}) = -\sqrt{5}$. The degree-three hsop in Theorem 2.8 gives a map $F_3 : V \to V^\sigma$. Let $F_5 : V \to V$ and $G_5 : V \to V^\sigma$ be the two quintic maps in Lemma 4.5. Applying σ to their coefficients gives $F_5^\sigma : V^\sigma \to V^\sigma$ and $G_5^\sigma : V^\sigma \to V$, since σ^2 fixes K_{H_3}. Thus $F_5^\sigma \circ F_3$ and $G_5^\sigma \circ F_3$ are equivariant maps of degree 15, with targets V^σ and V, respectively. Each composite vanishes only at the origin by Lemma 2.9. Their coordinate spans therefore realize both $(15, (V^*)^\sigma)$ and $(15, V^*)$.

Both quotients have dimension $15^3 = 3375$ by Proposition 2.3; the untwisted quotient is a permutation module and the twisted quotient is not, by Theorem 1.8.

4.4.2. *Verification of the seeds.*

Proof of Lemma 4.5. The cubic seeds for G_4 and G_6. For G_4, G_6 the maps are

W	k	$F(x,y)$
G_4	1	$(x^3+4y^3, -3x^2y)$
G_4	2	$(x^3-2y^3, -3xy^2)$
G_6	7	$(x^3-4y^3, -3x^2y)$
G_6	11	$(x^3+2y^3, 3xy^2)$.

Their only common zero is the origin. For these maps and the two G_{27} maps below, we use source generators $g = M_g^{-\mathsf{T}}$ from CHEVIE's matrices.

The quintic seeds for H_3. For H_3, put $a^2 = 5$, $\phi := (1+a)/2$, and use the generators

$$s_1 := \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad s_2 := \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad s_3 := \frac{1}{4}\begin{pmatrix} 2 & -1-a & 1-a \\ -1-a & 1-a & -2 \\ 1-a & -2 & 1+a \end{pmatrix}.$$

They preserve the 30 roots consisting of $\pm e_i$ and the cyclic permutations of $\frac{1}{2}(\pm 1, \pm\phi, \pm(\phi-1))$, and generate a group of order 120. Define

$$f := x\big(x^4 + bz^4 + cx^2y^2 + dx^2z^2 + ey^2z^2\big), \qquad F := (f(x,y,z), f(y,z,x), f(z,x,y)).$$

Use the coefficients

$\sigma(a)$	b	c	d	e
a	$(a+5)/2$	$a+5$	0	$-2a$
$-a$	$-3a$	$(3a+5)/2$	-5	$(9a-15)/2$.

Direct substitution proves $F(gv) = \sigma(g)F(v)$ for the three generators. The nontrivial twist can be represented by either $k = 2$ or $k = 3$ modulo 5.

The quintic seeds for G_{27}. For G_{27}, in the same matrix convention, take $F_2 := (f_1, f_2, f_3)$, where

$$\begin{aligned} f_1 &:= x^5 + 2x^3y^2 - 3xy^4 - 6x^3yz + 6xy^3z - 3x^3z^2 - 9xy^2z^2 - 18xyz^3, \\ f_2 &:= -3x^4y + 2x^2y^3 + y^5 - 3x^4z + 3y^4z + 9x^2yz^2 + 3y^3z^2 - 9x^2z^3 - 9y^2z^3, \\ f_3 &:= -2x^4y - 4x^2y^3 - \tfrac{2}{5}y^5 + 3x^4z - 6x^2y^2z + 3y^4z - 3x^2z^3 + 3y^2z^3 + \tfrac{18}{5}z^5. \end{aligned}$$

For the second twist take $F_{11} := (g_1, g_2, g_3)$, where

$$\begin{aligned} g_1 &:= x^5 - 2x^3y^2 + xy^4 + 6x^3yz + 6xy^3z + 3x^3z^2 - 3xy^2z^2 - \tfrac{9}{2}xz^4, \\ g_2 &:= x^4y - 2x^2y^3 + y^5 - \tfrac{3}{2}x^4z - 9x^2y^2z - \tfrac{3}{2}y^4z + 3x^2yz^2 - 3y^3z^2 - \tfrac{9}{2}yz^4, \\ g_3 &:= -x^4y - 2x^2y^3 - \tfrac{1}{5}y^5 - x^4z + 2x^2y^2z - y^4z + 6x^2z^3 - 6y^2z^3 + \tfrac{9}{5}z^5. \end{aligned}$$

The coordinates lie in $\mathbb{Q}[x,y,z]$, although their covariance uses the nontrivial actions of σ_2 and σ_{11} on $\mathbb{Q}(\zeta_{15})$.

For all eight maps, the package [41] checks homogeneity and verifies $F(gv) = \sigma(g)F(v)$ for every source generator. The calculations use $\mathbb{Q}(a)$ for H_3 and the corresponding cyclotomic fields for G_4, G_6, G_{27}. Gröbner bases of the coordinate ideals give quotient dimensions 9 for the four cubics and 125 for the four quintics. These quotients are finite-dimensional, so homogeneity implies that each map vanishes only at the origin. For H_3, the package also verifies the 30 roots and group order 120. For the six cyclotomic maps, it also compares the printed coefficients with the stored data and retains the covariant-space searches that produced the maps. □

4.5. **Restriction constructions.** The remaining non-well-generated exceptional groups follow from two restriction arguments. In rank two we extend known hsop maps to scalar overgroups; for G_{31} we restrict an E_8 map to a regular eigenspace.

4.5.1. *The non-well-generated rank-two groups.* We extend the covariance of known hsop maps by scalar homogeneity, then restrict to the seven non-well-generated rank-two groups.

Lemma 4.8. *Let $H \subseteq \mathrm{GL}(V)$ be an irreducible well-generated complex reflection group with Coxeter number h, and let M be a positive multiple of h. If $p > 0$ and $\gcd(p, M) = 1$, then every subgroup $W \subseteq \mu_M H$ has a degree-p hsop carrying $(V^*)^{\sigma_p}$, where $\sigma_p(\zeta_M) = \zeta_M^p$.*

Proof. Choose the degree-p map from Theorem 2.8. Homogeneity gives $F(\lambda v) = \lambda^p F(v) = \sigma_p(\lambda)F(v)$ for $\lambda \in \mu_M$. Thus its covariance extends from H to $\mu_M H$, and then restricts to W. Its zero locus is unchanged. □

Apply the lemma to the following groups:

(9)

W	H	M	common projective image
G_7	G_4	12	A_4
$G_{11}, G_{12}, G_{13}, G_{15}$	G_8	24	S_4
G_{19}, G_{22}	G_{16}	60	A_5.

Here the required inclusions follow directly from the classification of primitive finite subgroups of $\mathrm{GL}_2(\mathbb{C})$ by Shephard–Todd [38]. After conjugating their projective images to agree, write each $w \in W$ as λh with $h \in H$. In each row the determinant characters of both groups have order dividing $M/2$. Therefore $\lambda^2 = \det(w)/\det(h) \in \mu_{M/2}$ and $\lambda \in \mu_M$.

For each group in Eq. (9), the calculation in Proposition 4.4 shows that condition (3) of Theorem 1.6 is equivalent to $\gcd(p, M) = 1$ and $U \cong (V^*)^{\sigma_p}$. Lemma 4.8 therefore proves sufficiency in every degree. The package [41] checks the determinant and projective orders and compares the degree residues of posh pairs over a full period; the inclusions themselves are justified by the preceding argument.

4.5.2. *Restriction from E_8 to G_{31}.* The last exceptional group is obtained by restricting an E_8 hsop to a regular eigenspace. Homogeneity determines the target eigenspace and therefore its Galois twist.

Lemma 4.9. *Suppose a finite group $\widetilde{W}$ acts on a rational vector space $\widetilde{V}_{\mathbb{Q}}$, and put $\widetilde{V} \coloneqq \mathbb{C} \otimes_{\mathbb{Q}} \widetilde{V}_{\mathbb{Q}}$. Let $g \in \widetilde{W}$ have a root-of-unity eigenvalue ζ on $\widetilde{V}$, and put $V \coloneqq \ker(g - \zeta)$ and $W \coloneqq C_{\widetilde{W}}(g)$. If $p > 0$ and $\gcd(p, \operatorname{ord}\zeta) = 1$, then a homogeneous degree-p $\widetilde{W}$-equivariant map $F : \widetilde{V} \to \widetilde{V}$ with $F^{-1}(0) = \{0\}$ restricts to*

$$V \longrightarrow \ker(g - \zeta^p) \cong V^{\sigma_p}, \qquad \sigma_p(\zeta) = \zeta^p,$$

and its coordinate functions span an hsop for the action of W on V.

Proof. For $v \in V$, equivariance and homogeneity give $gF(v) = F(gv) = \zeta^p F(v)$. The restricted map has no nonzero zero. Rationality identifies the two eigenspaces by Galois conjugation, with equal dimension and the asserted action of the centralizer. The height bound then forces the coordinate functions of the restricted map to be linearly independent, so their span has dimension $\dim V$. □

The group G_{31} is the centralizer of a regular element of order 4 in E_8, acting on its i-eigenspace [9, Theorem 1.9(2) and Example 1.10]. The invariant degrees on that eigenspace are $8, 12, 20, 24$. For every p coprime to 30, the rational reflection representation of E_8 has an untwisted degree-p hsop by Theorem 2.8. Since p is odd, Lemma 4.9 gives a G_{31} hsop with trivial twist if $p \equiv 1 \pmod 4$, and conjugate twist if $p \equiv 3 \pmod 4$. These are precisely the two generator types and their degrees for G_{31} in Table 5.

The package [41] checks the degree selection by divisibility by four and the two twist congruences. This is an arithmetic check of the restriction argument.

4.6. **Completion and specializations.**

Completion of the proof of Theorem 1.6. Proposition 4.1 proves (1) ⇒ (2) ⇒ (3) and reduces (3) to Galois twists of V^*. Propositions 4.2 and 4.4 then give (3) ⇒ (4).

For (4) ⇒ (1), Proposition 4.2 proves that the listed infinite-family pairs are posh. For well-generated exceptional groups with $\gcd(p, h) = 1$, the listed type is $U \cong (V^*)^{\sigma_p}$ and Theorem 2.8 supplies the hsop. Proposition 4.6 treats all remaining pairs for G_4, G_6, H_3, G_{27}. Finally, the applications of Lemmas 4.8 and 4.9 in Section 4.5 give every listed pair for the non-well-generated exceptional groups. These cases exhaust the tables and complete the proof. □

Proposition 1.5. *Let W be a well-generated finite irreducible complex reflection group, let $p > 0$ be coprime to its Coxeter number h, and let U be a W-module. Then (p, U) is posh iff $U \simeq (V^*)^{\sigma_p}$.*

Proof. Theorem 2.8 proves that $(p, (V^*)^{\sigma_p})$ is posh. Conversely, the well-generated rows of Tables 3 and 5 in Theorem 1.6 allow only $U \cong (V^*)^{\sigma_p}$ when $\gcd(p, h) = 1$. □

Corollary 1.9. *Let W be an irreducible well-generated complex reflection group with Coxeter element c of order h, and let Θ be a posh hsop of degree p with $\gcd(p, h) = 1$. Write $e_i \coloneqq d_i - 1$ for the exponents of W.*

The following are equivalent:

- *Θ carries V^*;*

- A_Θ *is a permutation posh parking space;*
- c^p *is conjugate to* c *in* W*; and*
- $\{pe_i \bmod h\}_{i=1}^r = \{e_i\}_{i=1}^r$ *as multisets, where residues are taken in* $\{0, \dots, h-1\}$.

Proof. Let σ_p be the automorphism of $\mathbb{Q}(\zeta_h)$ defined by $\sigma_p(\zeta_h) = \zeta_h^p$. By Proposition 1.5, $\Theta \cong (V^*)^{\sigma_p}$, so Theorem 1.8 gives

$$A_\Theta \text{ is a permutation module} \iff (V^*)^{\sigma_p} \cong V^* \iff \sigma_p|_{K_W} = \mathrm{id}.$$

The last equivalence follows because the character values of V^* generate K_W. By V. Reiner, V. Ripoll, and C. Stump [36, Theorem 1.8(iv)], the coefficients of the characteristic polynomial of c on V also generate K_W. Since c has order h, applying σ_p to these coefficients gives the characteristic polynomial of c^p. Thus $\sigma_p|_{K_W} = \mathrm{id}$ exactly when c and c^p have the same eigenvalues, with multiplicities. Because $\gcd(p, h) = 1$, both are regular of order h, and this equality is equivalent to conjugacy in W by Reiner–Ripoll–Stump [36, Lemma 3.4 and its proof].

Choose a primitive hth root ζ for which c is ζ-regular. By Springer's eigenvalue theorem, as recalled in Reiner–Ripoll–Stump [36, Theorem 2.4(ii)], the eigenvalues of c on V are $\zeta^{-e_1}, \dots, \zeta^{-e_r}$. Thus c^p has the same eigenvalues exactly when the multisets of residues pe_i and e_i modulo h agree. Since $0 \le e_i < h$, this is the displayed criterion. □

Specializing the classification to $U = V^*$ and applying Theorem 1.8 gives Tables 4 and 6. In the infinite families the degrees are $\gcd(p, n) = 1$ for S_n, $p \equiv 1 \pmod m$ for non-dihedral $G(m, d, r)$, and $p \equiv \pm 1 \pmod m$ for $I_2(m)$.

5. Further questions

5.1. **Capricious hsops.** It seems natural to ask what happens if we relax the conditions in Definition 1.1 to allow a graded W-stable subspace $\Theta = \bigoplus_{a>0} \Theta_a \subseteq S$ of dimension r, and require $S/(\Theta)$ to be finite-dimensional. We call such a Θ a *capricious hsop*, and its quotient $S/(\Theta)$ a *capricious parking space* [25, Section 6, p. 1274].

A capricious hsop may be a nonfaithful representation of W, and its homogeneous generators may have different degrees. For irreducible W, posh hsops are precisely the faithful capricious hsops: the height argument of Theorem 3.4 also applies to generators of different degrees, so Proposition 2.6 makes a faithful Θ a Galois twist of V^*; a graded copy must lie in one degree.

Problem 5.1. For an irreducible complex reflection group W, determine which nonfaithful graded W-modules occur as capricious hsops, and characterize their possible degrees.

Several constructions illustrate the importance of this extension.

- Let $g_1, \dots, g_r \in S^W$ be homogeneous invariant parameters of degrees $a_1, \dots, a_r$, and put $B \coloneqq S^W/(g_1, \dots, g_r)$. Tensoring the decomposition in Eq. (1) with B gives

  $$S/(g_1, \dots, g_r) \cong C_W \otimes B, \qquad \mathrm{Hilb}(B; q) = \prod_{i=1}^r \frac{[a_i]_q}{[d_i]_q} \in \mathbb{Z}_{\ge 0}[q].$$

  The coefficient of q^j in $\mathrm{Hilb}(B; q)$ counts copies of C_W whose grading is shifted by j. After forgetting grading, the quotient is a sum of $\prod_i a_i/|W|$ regular representations. The construction in Proposition 3.2 supplies such capricious hsops of degrees $a_i = p + e_i(\Theta^*)$ from any W-stable hsop $\Theta \subseteq S_p$.
- Taking $g_i = f_i$, the fundamental invariants, gives $B = \mathbb{C}$ and recovers the coinvariant algebra $C_W = S/(f_1, \dots, f_r)$ in Section 2.1. The regular action of W on itself gives an ungraded combinatorial model for this quotient.
- If W is irreducible and well-generated, choose the top degree invariant ($\deg f_r = h$) so that the reduced branch discriminant is monic and centered in f_r, say

  $$\Delta(f_1, \dots, f_r) = f_r^r + \alpha_2 f_r^{r-2} + \cdots + \alpha_r, \qquad \alpha_i \in \mathbb{C}[f_1, \dots, f_{r-1}].$$

  Finiteness of the Lyashko–Looijenga map makes $f_r, \alpha_2, \dots, \alpha_r$ invariant parameters of degrees $h, 2h, \dots, rh$; see [42, Section 7.2] and [13, Lemma 4.4]. Here $\mathrm{Hilb}(B; q) = \prod_{i=1}^r [ih]_q/[d_i]_q$. The invariant subspace of the quotient has dimension $r!h^r/|W|$, the number of reduced reflection factorizations of a Coxeter element c. These factorizations index a generic fiber of the Lyashko–Looijenga map [9, Theorem 7.20].

- A final motivation comes from Griffeth's work on diagonal coinvariants. $V \oplus V^*$ is the underlying complex representation of the quaternionification of V, and the diagonal coinvariant ring
$$D_W \coloneqq \mathbb{C}[V \oplus V^*] \big/ \big\langle \big(\mathbb{C}[V \oplus V^*]^W\big)_+ \big\rangle$$
records the scheme-theoretic zero fiber of the quotient by W. For irreducible W, put $g = 2N/r$, where N is the number of reflections; this is an integer. Griffeth's construction from rational Cherednik algebras gives an hsop of degree $g+1$ carrying the irreducible KZ twist $\kappa(V^*)$, where κ is the representation permutation in that construction [21, Theorem 1.2 and Section 3R]. Its quotient L has dimension $(g+1)^r$. After tensoring by $\det(V)^*$, the associated graded module of L for the filtration generated by its unique determinant copy is a quotient of D_W [21, Lemma 3.1]. These generators need not be faithful: for $W = G(6,3,r)$, $r \geq 2$, the degree is $6r-3$ and
$$\kappa(V^*) \cong \operatorname{span}_{\mathbb{C}}\{x_1^{6r-3}, \dots, x_r^{6r-3}\}, \qquad \ker \kappa(V^*) \cong (\mathbb{Z}/3\mathbb{Z})^{r-1};$$
see [42, Section 8].

5.2. **Polynomial differential forms.** A second generalization retains the posh condition but adds differential information. For a posh hsop $\Theta \subseteq S_p$, define
$$\Omega_\Theta \coloneqq \big(S \otimes \textstyle\bigwedge V^*\big)/(\Theta, d\Theta), \qquad d\theta = \sum_{i=1}^r \frac{\partial \theta}{\partial x_i}\, dx_i.$$
Here $x_1, \dots, x_r$ is a basis of V^*, and the exterior generators dx_i transform in another copy of V^*. Give x_i bidegree $(1,0)$ and dx_i bidegree $(0,1)$, so that θ and $d\theta$ have bidegrees $(p,0)$ and $(p-1,1)$, respectively. The relations $d\theta = 0$ give the Kähler differentials of A_Θ [43, Tag 00RU]: a differential form on the quotient must respect the defining relations $\theta = 0$. Thus
$$\Omega_\Theta \cong \textstyle\bigwedge_{A_\Theta} \Omega^1_{A_\Theta/\mathbb{C}}.$$
Its component of exterior degree zero is A_Θ. The same construction for the capricious hsop of fundamental invariants gives the usual super coinvariant algebra [45, Section 5].

Problem 5.2. For a fixed posh pair (p, U), determine the bigraded W-characters of Ω_Θ as Θ varies over the degree-p posh hsops carrying U. Characterize those Θ for which Ω_Θ, after forgetting both gradings, is a permutation representation.

Unlike the character of A_Θ, the character of Ω_Θ can depend on the choice of hsop within a fixed posh pair.

Example 5.3. Let $W = B_2$ act on $\mathbb{C}[x,y]$ by signed permutations, and put
$$\Theta_a \coloneqq \operatorname{span}_{\mathbb{C}}\{x^3 + axy^2,\ y^3 + ax^2y\}, \qquad a \in \mathbb{C} \setminus \{1, -1\}.$$
These are posh hsops carrying V^*. Put $\Omega_a \coloneqq \Omega_{\Theta_a}$, and write Ω_a^j for its component of exterior degree j. Its top component is
$$\Omega_a^2 \cong \frac{\mathbb{C}[x,y]}{(3x^2 + ay^2,\ 2axy,\ 3y^2 + ax^2)}\, dx \wedge dy.$$
Euler's identity shows that the cubic generators already lie in the displayed ideal. For $a \notin \{0, \pm 3\}$, the displayed quotient of $\mathbb{C}[x,y]$ has basis $1, x, y$. At $a = 0, 3, -3$, it has one additional class represented by xy, $x^2 - y^2$, and $x^2 + y^2$, respectively.

The exterior derivative and contraction with the Euler vector field $x\partial/\partial x + y\partial/\partial y$ descend to Ω_a: the derivative sends θ to $d\theta$ and kills $d\theta$, whereas contraction kills θ and sends $d\theta$ to 3θ. Both commute with W. The sum of their composites in the two orders acts on bidegree (k,j) by multiplication by $k+j$. The resulting de Rham complex is therefore exact in positive total degree and has only $\mathbb{C}$ in total degree zero. Since $\Omega_a^0 = A_{\Theta_a}$, its alternating character gives
$$[\Omega_a] = 2[A_{\Theta_a}] + 2[\Omega_a^2] - [\mathbb{C}]$$
in the representation ring of W, after forgetting both gradings. Combining the signed-permutation actions on the displayed basis classes, including the determinant action on $dx \wedge dy$, with the quotient character in Proposition 2.3 gives the following table. Let c be a quarter-turn, s an axis reflection, and t a diagonal reflection. The ordinary ungraded characters are

a	1	c^2	c	s	t
$a \notin \{0, \pm 1, \pm 3\}$	23	-1	3	3	3
0	25	1	1	5	1
3	25	1	1	1	5
-3	25	1	5	1	1

Ω_a is a permutation module exactly for $a = 0$ and $a = 3$. After forgetting both gradings,

$$\Omega_0 \cong \mathbb{C} \oplus \mathbb{C}[W]^{\oplus 2} \oplus \mathbb{C}[W/\langle s \rangle]^{\oplus 2},$$
$$\Omega_3 \cong \mathbb{C} \oplus \mathbb{C}[W]^{\oplus 2} \oplus \mathbb{C}[W/\langle t \rangle]^{\oplus 2}.$$

For all other admissible parameters, a permutation character is impossible: the generic character takes value -1 at c^2, and at $a = -3$ its values 5 at c and 1 at c^2 contradict $\mathrm{Fix}(c) \subseteq \mathrm{Fix}(c^2)$ for a permutation action.

Acknowledgements

We thank Minh-Tâm Trinh, Stephen Griffeth, Theo Douvropoulos, and Carlos Arreche for many helpful discussions and suggestions, and the University of Texas at Dallas for providing inspiration through its fashionable multi-tiered parking permit system. Both authors were partially supported by NSF grant DMS-2246877. We are grateful to OpenAI for support through an Academic Workspace; although our original code from 2024 constructing hsops was written by hand in Mathematica and GAP3, for this paper we used various GPT models for exploration, for computation, and for writing. We take full responsibility for all wordplay.

Appendix A. Classification tables

The tables list all posh pairs and their untwisted specializations from Theorems 1.6 and 1.8. Recall that p is a positive integer, $\mathfrak{f}_W$ is the conductor of the reflection character field, and k specifies the automorphism $\sigma_k(\zeta_{\mathfrak{f}_W}) = \zeta_{\mathfrak{f}_W}^k$ (where $\zeta_{\mathfrak{f}_W} := \exp(2\pi i/\mathfrak{f}_W)$ and $\gcd(k, \mathfrak{f}_W) = 1$). We list one k for each distinct twisted character. The integer M is the least period of the degrees p for which $(p, (V^*)^{\sigma_k})$ is posh: an entry R in the column $p \bmod M$ means exactly the positive degrees $p \equiv s \pmod M$ with $s \in R$.

Group	Family parameters	Degree p and module U	Construction
S_n	$n \geq 2$, $r = n - 1$	$\gcd(p, n) = 1$, $U \cong V^*$	Example 2.11
$G(m, d, r)$	$m \geq 2$, $d \mid m$, $r \geq 1$; $d = 1$ if $r = 1$; $(m, d, r) \neq (2, 2, 2)$	$\gcd(p, m) = 1$, $U \cong (V^*)^{\sigma_p}$	Section 4.2

Table 3. Posh pairs in the infinite families, for Theorem 1.6.

Group	Family parameters	Degree p	Construction
S_n	$n \geq 2$, $r = n - 1$	$\gcd(p, n) = 1$	Example 2.11
$G(m, d, r)$	$m \geq 2$, $d \mid m$, $r \geq 1$; $d = 1$ if $r = 1$; $(d, r) \neq (m, 2)$	$p \equiv 1 \pmod m$	Section 4.2
$G(m, m, 2)$	$m \geq 3$, $r = 2$	$p \equiv \pm 1 \pmod m$	Section 4.2

Table 4. Untwisted posh pairs (p, V^*) in the infinite families, for Theorem 1.8. The degree conditions agree with [24, Table 1].

W	Degrees	h	$\mathfrak{f}_W$	k	Period M	$p \bmod M$	Construction
G_4	$4, 6$	6	3	1	6	$\{1, 3\}$	Section 4.4.1
				2		$\{3, 5\}$	
G_5	$6, 12$	12	3	1	6	$\{1\}$	Theorem 2.8
				2		$\{5\}$	
G_6	$4, 12$	12	12	1	12	$\{1, 9\}$	Section 4.4.1
				5		$\{5, 9\}$	
				7		$\{3, 7\}$	
				11		$\{3, 11\}$	
G_7	$12, 12$	–	12	1	12	$\{1\}$	Lemma 4.8
				5		$\{5\}$	
				7		$\{7\}$	
				11		$\{11\}$	
G_8	$8, 12$	12	4	1	12	$\{1, 5\}$	Theorem 2.8
				3		$\{7, 11\}$	
G_9	$8, 24$	24	8	1	24	$\{1, 17\}$	Theorem 2.8
				3		$\{11, 19\}$	
				5		$\{5, 13\}$	
				7		$\{7, 23\}$	
G_{10}	$12, 24$	24	12	1	12	$\{1\}$	Theorem 2.8
				5		$\{5\}$	
				7		$\{7\}$	
				11		$\{11\}$	
G_{11}	$24, 24$	–	24	1	24	$\{1\}$	Lemma 4.8
				5		$\{5\}$	
				7		$\{7\}$	
				11		$\{11\}$	
				13		$\{13\}$	
				17		$\{17\}$	
				19		$\{19\}$	
				23		$\{23\}$	
G_{12}	$6, 8$	–	8	1	24	$\{1, 11, 17, 19\}$	Lemma 4.8
				5		$\{5, 7, 13, 23\}$	
G_{13}	$8, 12$	–	8	1	24	$\{1, 17\}$	Lemma 4.8
				3		$\{11, 19\}$	
				5		$\{5, 13\}$	
				7		$\{7, 23\}$	
G_{14}	$6, 24$	24	24	1	24	$\{1, 19\}$	Theorem 2.8
				5		$\{5, 23\}$	
				7		$\{7, 13\}$	
				11		$\{11, 17\}$	
G_{15}	$12, 24$	–	24	1	24	$\{1\}$	Lemma 4.8
				5		$\{5\}$	
				7		$\{7\}$	
				11		$\{11\}$	
				13		$\{13\}$	
				17		$\{17\}$	
				19		$\{19\}$	
				23		$\{23\}$	
G_{16}	$20, 30$	30	5	1	30	$\{1, 11\}$	Theorem 2.8
				2		$\{7, 17\}$	
				3		$\{13, 23\}$	
				4		$\{19, 29\}$	

W	Degrees	h	$\mathfrak{f}_W$	k	Period M	$p \bmod M$	Construction
G_{17}	$20, 60$	60	20	1	60	$\{1, 41\}$	Theorem 2.8
				3		$\{23, 43\}$	
				7		$\{7, 47\}$	
				9		$\{29, 49\}$	
				11		$\{11, 31\}$	
				13		$\{13, 53\}$	
				17		$\{17, 37\}$	
				19		$\{19, 59\}$	
G_{18}	$30, 60$	60	15	1	30	$\{1\}$	Theorem 2.8
				2		$\{17\}$	
				4		$\{19\}$	
				7		$\{7\}$	
				8		$\{23\}$	
				11		$\{11\}$	
				13		$\{13\}$	
				14		$\{29\}$	
G_{19}	$60, 60$	–	60	1	60	$\{1\}$	Lemma 4.8
				7		$\{7\}$	
				11		$\{11\}$	
				13		$\{13\}$	
				17		$\{17\}$	
				19		$\{19\}$	
				23		$\{23\}$	
				29		$\{29\}$	
				31		$\{31\}$	
				37		$\{37\}$	
				41		$\{41\}$	
				43		$\{43\}$	
				47		$\{47\}$	
				49		$\{49\}$	
				53		$\{53\}$	
				59		$\{59\}$	
G_{20}	$12, 30$	30	15	1	30	$\{1, 19\}$	Theorem 2.8
				2		$\{17, 23\}$	
				7		$\{7, 13\}$	
				11		$\{11, 29\}$	
G_{21}	$12, 60$	60	60	1	60	$\{1, 49\}$	Theorem 2.8
				7		$\{7, 43\}$	
				11		$\{11, 59\}$	
				13		$\{13, 37\}$	
				17		$\{17, 53\}$	
				19		$\{19, 31\}$	
				23		$\{23, 47\}$	
				29		$\{29, 41\}$	
G_{22}	$12, 20$	–	20	1	60	$\{1, 29, 41, 49\}$	Lemma 4.8
				3		$\{7, 23, 43, 47\}$	
				11		$\{11, 19, 31, 59\}$	
				13		$\{13, 17, 37, 53\}$	
G_{23}	$2, 6, 10$	10	5	1	10	$\{1, 5, 9\}$	Section 4.4.1
				2		$\{3, 5, 7\}$	
G_{24}	$4, 6, 14$	14	7	1	14	$\{1, 9, 11\}$	Theorem 2.8
				3		$\{3, 5, 13\}$	
G_{25}	$6, 9, 12$	12	3	1	6	$\{1\}$	Theorem 2.8
				2		$\{5\}$	
G_{26}	$6, 12, 18$	18	3	1	6	$\{1\}$	Theorem 2.8
				2		$\{5\}$	

W	Degrees	h	$\mathfrak{f}_W$	k	Period M	$p \bmod M$	Construction
G_{27}	$6, 12, 30$	30	15	1	30	$\{1, 19, 25\}$	Section 4.4.1
				2		$\{5, 17, 23\}$	
				7		$\{7, 13, 25\}$	
				11		$\{5, 11, 29\}$	
G_{28}	$2, 6, 8, 12$	12	1	1	6	$\{1, 5\}$	Theorem 2.8
G_{29}	$4, 8, 12, 20$	20	4	1	20	$\{1, 9, 13, 17\}$	Theorem 2.8
				3		$\{3, 7, 11, 19\}$	
G_{30}	$2, 12, 20, 30$	30	5	1	30	$\{1, 11, 19, 29\}$	Theorem 2.8
				2		$\{7, 13, 17, 23\}$	
G_{31}	$8, 12, 20, 24$	–	4	1	60	$\{1, 13, 17, 29, 37, 41, 49, 53\}$	Lemma 4.9
				3		$\{7, 11, 19, 23, 31, 43, 47, 59\}$	
G_{32}	$12, 18, 24, 30$	30	3	1	30	$\{1, 7, 13, 19\}$	Theorem 2.8
				2		$\{11, 17, 23, 29\}$	
G_{33}	$4, 6, 10, 12, 18$	18	3	1	6	$\{1\}$	Theorem 2.8
				2		$\{5\}$	
G_{34}	$6, 12, 18, 24, 30, 42$	42	3	1	42	$\{1, 13, 19, 25, 31, 37\}$	Theorem 2.8
				2		$\{5, 11, 17, 23, 29, 41\}$	
G_{35}	$2, 5, 6, 8, 9, 12$	12	1	1	6	$\{1, 5\}$	Theorem 2.8
G_{36}	$2, 6, 8, 10, 12, 14, 18$	18	1	1	6	$\{1, 5\}$	Theorem 2.8
G_{37}	$2, 8, 12, 14, 18, 20, 24, 30$	30	1	1	30	$\{1, 7, 11, 13, 17, 19, 23, 29\}$	Theorem 2.8

Table 5: Exceptional posh pairs in Theorem 1.6, with $U = (V^*)^{\sigma_k}$. The Coxeter number $h = d_r$ is displayed precisely for the well-generated groups. Only $k = 1$ gives a permutation module.

W	Degrees	h	Period M	$p \bmod M$
G_4	$4, 6$	6	6	$\{1, 3\}$
G_5	$6, 12$	12	6	$\{1\}$
G_6	$4, 12$	12	12	$\{1, 9\}$
G_7	$12, 12$	–	12	$\{1\}$
G_8	$8, 12$	12	12	$\{1, 5\}$
G_9	$8, 24$	24	24	$\{1, 17\}$
G_{10}	$12, 24$	24	12	$\{1\}$
G_{11}	$24, 24$	–	24	$\{1\}$
G_{12}	$6, 8$	–	24	$\{1, 11, 17, 19\}$
G_{13}	$8, 12$	–	24	$\{1, 17\}$
G_{14}	$6, 24$	24	24	$\{1, 19\}$
G_{15}	$12, 24$	–	24	$\{1\}$
G_{16}	$20, 30$	30	30	$\{1, 11\}$
G_{17}	$20, 60$	60	60	$\{1, 41\}$
G_{18}	$30, 60$	60	30	$\{1\}$
G_{19}	$60, 60$	–	60	$\{1\}$
G_{20}	$12, 30$	30	30	$\{1, 19\}$
G_{21}	$12, 60$	60	60	$\{1, 49\}$
G_{22}	$12, 20$	–	60	$\{1, 29, 41, 49\}$
$H_3 = G_{23}$	$2, 6, 10$	10	10	$\{1, 5, 9\}$
G_{24}	$4, 6, 14$	14	14	$\{1, 9, 11\}$
G_{25}	$6, 9, 12$	12	6	$\{1\}$
G_{26}	$6, 12, 18$	18	6	$\{1\}$
G_{27}	$6, 12, 30$	30	30	$\{1, 19, 25\}$
$F_4 = G_{28}$	$2, 6, 8, 12$	12	6	$\{1, 5\}$
G_{29}	$4, 8, 12, 20$	20	20	$\{1, 9, 13, 17\}$
$H_4 = G_{30}$	$2, 12, 20, 30$	30	30	$\{1, 11, 19, 29\}$
G_{31}	$8, 12, 20, 24$	–	60	$\{1, 13, 17, 29, 37, 41, 49, 53\}$
G_{32}	$12, 18, 24, 30$	30	30	$\{1, 7, 13, 19\}$
G_{33}	$4, 6, 10, 12, 18$	18	6	$\{1\}$
G_{34}	$6, 12, 18, 24, 30, 42$	42	42	$\{1, 13, 19, 25, 31, 37\}$
$E_6 = G_{35}$	$2, 5, 6, 8, 9, 12$	12	6	$\{1, 5\}$
$E_7 = G_{36}$	$2, 6, 8, 10, 12, 14, 18$	18	6	$\{1, 5\}$
$E_8 = G_{37}$	$2, 8, 12, 14, 18, 20, 24, 30$	30	30	$\{1, 7, 11, 13, 17, 19, 23, 29\}$

TABLE 6. Untwisted exceptional posh pairs (p, V^*) in Theorem 1.8. These are the $k = 1$ rows of Table 5; the Coxeter number $h = d_r$ is displayed precisely for the well-generated groups. The degree conditions agree with [24, Table 1].

References

[1] D. Armstrong, N. A. Loehr, and G. S. Warrington, *Rational parking functions and Catalan numbers*, Ann. Comb. **20** (2016), no. 1, 21–58, doi:10.1007/s00026-015-0293-6, arXiv:1403.1845.

[2] D. Armstrong, V. Reiner, and B. Rhoades, *Parking spaces*, Adv. Math. **269** (2015), 647–706, arXiv:1204.1760.

[3] C. E. Arreche and N. F. Williams, *Normal reflection subgroups of complex reflection groups*, J. Inst. Math. Jussieu **22** (2023), no. 2, 879–917, doi:10.1017/S1474748021000323, arXiv:2007.09778.

[4] M. Balagović and A. Puranik, *Irreducible representations of the rational Cherednik algebra associated to the Coxeter group H_3*, J. Algebra **405** (2014), 259–290, arXiv:1004.2108.

[5] V. Beck, *Exterior algebra structure on relative invariants of reflection groups*, Math. Z. **267** (2011), 261–289, doi:10.1007/s00209-009-0619-3.

[6] M. Benard, *Schur indices and splitting fields of the unitary reflection groups*, J. Algebra **38** (1976), no. 2, 318–342, doi:10.1016/0021-8693(76)90223-4.

[7] Y. Berest, P. Etingof, and V. Ginzburg, *Finite-dimensional representations of rational Cherednik algebras*, Int. Math. Res. Not. (2003), no. 19, 1053–1088, doi:10.1155/S1073792803210205, arXiv:math/0208138.

[8] D. Bessis, *Sur le corps de définition d'un groupe de réflexions complexe*, Comm. Algebra **25** (1997), no. 8, 2703–2716, doi:10.1080/00927879708826016.

[9] D. Bessis, *Finite complex reflection arrangements are $K(\pi, 1)$*, Ann. of Math. **181** (2015), 809–904. arXiv:math/0610777.

[10] D. Bessis and V. Reiner, *Cyclic sieving of noncrossing partitions for complex reflection groups*, Ann. Comb. **15** (2011), 197–222. arXiv:math/0701792.

[11] C. Chevalley, *Invariants of finite groups generated by reflections*, Amer. J. Math. **77** (1955), no. 4, 778–782.

[12] C. Defant and N. Williams, *Noncrossing combinatorics, the full twist, and decategorification of knot invariants*, arXiv:2608.06225.

[13] T. Douvropoulos, *Lyashko–Looijenga morphisms and primitive factorizations of the Coxeter element*, arXiv:1808.10395v2.

[14] T. Douvropoulos and M. Josuat-Vergès, *The generalized cluster complex: refined enumeration of faces and related parking spaces*, SIGMA **19** (2023), 069, 40 pp., doi:10.3842/SIGMA.2023.069, arXiv:2209.12540.

[15] P. H. Edelman, *Chain enumeration and non-crossing partitions*, Discrete Math. **31** (1980), no. 2, 171–180, doi:10.1016/0012-365X(80)90033-3.

[16] P. Etingof, *Proof of the Broué–Malle–Rouquier conjecture in characteristic zero (after I. Losev and I. Marin–G. Pfeiffer)*, Arnold Math. J. **3** (2017), 445–449, doi:10.1007/s40598-017-0069-7, arXiv:1606.08456.

[17] P. Galashin, T. Lam, M.-T. Trinh, and N. Williams, *Rational noncrossing Coxeter–Catalan combinatorics*, Proc. Lond. Math. Soc. (3) **129** (2024), no. 4, e12643, doi:10.1112/plms.12643, arXiv:2208.00121.

[18] M. Geck, G. Hiss, F. Lübeck, G. Malle, and G. Pfeiffer, *CHEVIE—A system for computing and processing generic character tables*, Appl. Algebra Engrg. Comm. Comput. **7** (1996), 175–210.

[19] I. Gordon, *On the quotient ring by diagonal invariants*, Invent. Math. **153** (2003), no. 3, 503–518, doi:10.1007/s00222-003-0296-5, arXiv:math/0208126.

[20] I. Gordon and S. Griffeth, *Catalan numbers for complex reflection groups*, Amer. J. Math. **134** (2012), 1491–1502. arXiv:0912.1578.

[21] S. Griffeth, *The diagonal coinvariant ring of a complex reflection group*, Algebra Number Theory **17** (2023), no. 11, 2033–2053, doi:10.2140/ant.2023.17.2033, arXiv:2112.01419.

[22] M. Haiman, *Conjectures on the quotient ring by diagonal invariants*, J. Algebraic Combin. **3** (1994), 17–76, author PDF.

[23] H. Hu, *On exterior powers of reflection representations, II*, Canad. J. Math., doi:10.4153/S0008414X25000215, arXiv:2401.08215.

[24] Y. Ito and S. Okada, *On the existence of generalized parking spaces for complex reflection groups*, arXiv:1508.06846.

[25] A. G. Konheim and B. Weiss, *An occupancy discipline and applications*, SIAM J. Appl. Math. **14** (1966), no. 6, 1266–1274, doi:10.1137/0114101.

[26] I. Marin, *The cubic Hecke algebra on at most 5 strands*, J. Pure Appl. Algebra **216** (2012), 2754–2782, doi:10.1016/j.jpaa.2012.04.013, arXiv:1110.6621.

[27] I. Marin, *The freeness conjecture for Hecke algebras of complex reflection groups, and the case of the Hessian group G_{26}*, J. Pure Appl. Algebra **218** (2014), 704–720, doi:10.1016/j.jpaa.2013.08.009, arXiv:1210.5632.

[28] I. Marin and J. Michel, *Automorphisms of complex reflection groups*, Represent. Theory **14** (2010), 747–788, doi:10.1090/S1088-4165-2010-00380-5, arXiv:math/0701266.

[29] I. Marin and G. Pfeiffer, *The BMR freeness conjecture for the 2-reflection groups*, Math. Comp. **86** (2017), 2005–2023, doi:10.1090/mcom/3142, arXiv:1411.4760.

[30] J. Michel, *The development version of the CHEVIE package of GAP3*, J. Algebra **435** (2015), 308–336, doi:10.1016/j.jalgebra.2015.03.031, arXiv:1310.7905.

[31] A. Oblomkov and Z. Yun, *Geometric representations of graded and rational Cherednik algebras*, Adv. Math. **292** (2016), 601–706, doi:10.1016/j.aim.2016.01.015.

[32] E. M. Opdam, *Complex reflection groups and fake degrees*, arXiv:math/9808026, Lemma 2.1.

[33] P. Orlik and L. Solomon, *Unitary reflection groups and cohomology*, Invent. Math. **59** (1980), no. 1, 77–94, doi:10.1007/BF01390316.

[34] P. Orlik and L. Solomon, *Arrangements defined by unitary reflection groups*, Math. Ann. **261** (1982), 339–357, doi:10.1007/BF01455455.

[35] P. Orlik and L. Solomon, *Coxeter arrangements*, Singularities, Part 2, Proc. Sympos. Pure Math. **40**, Amer. Math. Soc., Providence, RI, 1983, 269–291, doi:10.1090/pspum/040.2/713255.

[36] V. Reiner, V. Ripoll, and C. Stump, *On non-conjugate Coxeter elements in well-generated reflection groups*, Math. Z. **285** (2017), 1041–1062, doi:10.1007/s00209-016-1736-4, arXiv:1404.5522.

[37] B. Rhoades, *Parking structures: Fuss analogs*, J. Algebraic Combin. **40** (2014), 417–473, arXiv:1205.4293.
[38] G. C. Shephard and J. A. Todd, *Finite unitary reflection groups*, Canad. J. Math. **6** (1954), 274–304. doi:10.4153/CJM-1954-028-3.
[39] E. Sommers, *A family of affine Weyl group representations*, Transform. Groups **2** (1997), 375–390, doi:10.1007/BF01234541.
[40] J. Stack, *Parking spaces for complex reflection groups*, arXiv:2502.01970.
[41] J. Stack and N. Williams, *Computations for Algebraic parking spaces for complex reflection groups*, version 2026-09-24.2, https://personal.utdallas.edu/~nxw170830/docs/algebraic-hsops/.
[42] J. Stack and N. Williams, *Equivariant systems of parameters for reflection groups*, working preprint, September 2026.
[43] The Stacks Project Authors, *The Stacks Project*, 2026, https://stacks.math.columbia.edu.
[44] R. P. Stanley, *Parking functions and noncrossing partitions*, Electron. J. Combin. **4** (1997), no. 2, Research Paper R20, doi:10.37236/1335.
[45] J. P. Swanson and N. R. Wallach, *Harmonic differential forms for pseudo-reflection groups I. Semi-invariants*, J. Combin. Theory Ser. A **182** (2021), 105474, doi:10.1016/j.jcta.2021.105474, arXiv:2001.06076.